MIL-E-9426F(AS)
13 October 1978
SUPERSEDING
MIL-E-9426E(AS)
1 May 1974

MILITARY SPECIFICATION

ESCAPE SYSTEMS, REQUIREMENTS CONFORMANCE DEMONSTRATIONS AND PERFORMANCE TESTS FOR: GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes (a) the procedures to be used in demonstrating automated aircrew escape system compliance with the design, installation and performance requirements of MIL-S-18471 or MIL-A-23121, as applicable, and (b) the demonstration and test requirements which must be satisfied prior to the release of an escape system for installation and service use. In addition, this specification provides guidance for developing demonstration and test programs for automated aircrew escape systems which do not conform to MIL-S-18471 or MIL-A-23121.

1.2 Classification. The demonstrations and tests described herein shall be of the following types, as herein defined.

a. Component/subsystem service release/qualification test.

b. Aircrew station fit and compatibility test.

c. Environmental conditioning and windblast tests.

d. System design compliance tests.

e. Maintainability and reliability tests.

1.3 The demonstrations and tests, described herein shall be conducted after completion of the component/subsystem/system development program and shall utilize production systems or an equivalent approved by the Government procuring activity. Such equivalent shall assure the trial usage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

of proposed production procedures for evaluation as in pilot production lots; thus providing both (1) tests results meaningful in the context of the ultimate production configuration and (2) experience with and opportunity to correct and improve procedures to be employed in producing escape systems so as to ensure the maximum production quality.

1.4 Safety. Testing required herein involves many potential hazards requiring concerted efforts, particularly on the part of management in the area of planning for and executing the plans for tests, to assure safety of personnel, equipment and facilities.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment; General Specification for

MIL-S-18471 - System, Aircrew Automated Escape, Ejection Seat Type; General Specification for

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-A-23121 - Aircrew Environmental Escape and Survival Cockpit Capsule System; General Specification for

MIL-D-23615 - Design and Evaluation of Cartridge Actuated Devices

MIL-S-81018 - Survival Container, Aircraft Seat, with Oxygen; General Specification for

MIL-S-81040 - Survival Kit Containers, Aircraft Seat, Without Oxygen; General Specification for

MIL-D-81514 - Device, Restraint Harness Take-Up, Inertia-Locking, Powered-Retracting; General Specification for

MIL-A-81573 - Aircrew Escape System Descriptive and Performance Data Presentations, Requirements and Formats for; General Specification for

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-C-81590	- Cockpit Canopy System, Fixed Wing Single and Multiplace Fighter, Attack, and Trainer Aircraft; General Specification for
MIL-A-81973	- Aircrew Automated Escape Systems, Test Vehicles for Ground Track Performance Testing, Rocket and/or Jet Engine Propelled; General Specification for
MIL-D-81980	- Design and Evaluation of Signal Transmission Subsystems; General Specification for
MIL-S-83490	- Specifications, Types and Forms
MIL-A-85041	- Aircrew Escape Propulsion Systems; General Specification for
AR 46	- HERO (Hazards of Electromagnetic Radiation to Ordnance) Requirements for HERO Tests, Analysis Documentation
AR 72	- Aircrew Automated Escape System, Service Release Test Programs, Contractor Furnished Equipments and Government Furnished Equipments; Requirements for
AR 105	- Aircrew Automated Escape System In-process Design Conformance Inspection Articles, Standard Procedures for Evaluation of; Requirements for

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods

MILITARY

MIL-STD-442	- Aerospace Telemetry Standards
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-STD-810	- Environmental Test Methods
MIL-STD-831	- Test Reports, Preparation of
MIL-STD-858	- Testing Standard for Personnel Parachutes
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-888	- Testing Program for Emergency Escape Capsule Recovery Systems
MIL-STD-961	- Outline Of Forms And Instructions For The Preparation Of Specifications And Associated Documents
MIL-STD-2067	- Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

(Indices indicating the paragraphs in which the above listed specifications are cited begin on page 89.)

3. REQUIREMENTS

3.1 Components. The complete aircrew escape system shall be tested in accordance with the procedures specified herein, as applicable to the specific aircraft. In general, the components parts of the escape system shall include the capsule(s), ejection seat(s), canopy(ies) hatch(es) and removal system, recovery subsystems, restraint subsystems, firing control linkages, seat installation(s), sequencing system, survival system equipment, and such other operating devices as are applicable. Capsule systems shall include all actual equipment, or Government procuring activity approved simulations of proper weight, size and centers of gravity installed in production systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.1.1 Test articles. Unless otherwise specified by the Government procuring activity, the test articles for all demonstrations and tests shall include all components, as defined in 3.1, of the aircrew escape system. The test articles shall be manufactured and assembled in accordance with production design, techniques, processes, and procedures, including quality assurance procedures. All lots of parts, components, subassemblies, etc., shall be subjected to standard planned production/receiving quality assurance procedures as established for each part, component, subassembly, etc., to determine the acceptability of each lot. Prior to assembly of each escape system test article and after the parts, component, subassemblies, etc., have passed lot inspection and have been accepted, each part selected by normal escape system production assembly procedures shall be inspected fully and the data recorded upon worksheets for the serialized test article. There shall be no non-production special selection/fitting/modification of parts, components, subassemblies, etc., to obtain fit, required performance, or to improve some other characteristic. In the event a part, component, subassembly, etc., is determined to have a non-conforming characteristic, the contractor immediately shall notify the resident Government quality assurance representative and, if a critical nonconformity, the Government procuring activity, or its designee, concerning the part number of the part, the nature of the deficiency, the anticipated problems associated with the use of the non-conforming part, action recommended concerning the particular non-conforming part, and action to be taken to preclude recurrence of similar nonconformities in future accepted lots. The contractor shall take precautions to ensure that the test article receives no special care and, accordingly, is representative of articles which will be produced by the established production techniques, processes and procedures.

3.1.1.1 Test article configuration changes. Significant changes in aircrew escape system and/or component design/configuration resulting from the required tests will necessitate retesting using the revised escape system configuration.

3.1.2 Test article reuse.

3.1.2.1 Capsule type escape systems. At least ninety (90) days prior to commencement of testing the contractor shall provide to the Government procuring activity for approval a list of components/subsystems which the contractor considers likely to be reusable. The list shall include a detailed explanation describing for each part why the contractor considers reuse feasible. Failure of the Government procuring activity to reject or modify the list within thirty (30) days, shall constitute approval of the list. Prior to reuse, components shall be inspected and, if required, repaired/refurbished, to ensure proper functioning and their acceptability for use in the test. The Contractor shall bear full responsibility for any failures resulting from reuse of components/subsystems. Failure of a reused escape system part whether as a consequence of the stresses incurred in the previous usage or from an error in the inspection and/or reestablishment shall not constitute justification for classifying the test as a "no test".

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.1.2.2 Ejection seat type escape systems. No escape systems, or components thereof, may be reused in testing without the prior concurrence of the Government procuring activity. If the contractor has reason to believe that the test articles, or components thereof, might be reused successfully, the components shall be identified to the Government procuring activity and an explanation describing why the contractor considers reuse feasible shall be provided to the Government procuring activity at least sixty (60) days prior to commencement of testing for consideration. Failure of the Government procuring activity to reject or modify the list within thirty (30) days shall constitute approval of the list. Prior to reuse, components shall be inspected and, if required, repaired/refurbished/replaced, to ensure proper functioning during the test. The contractor shall bear full responsibility for any failures resulting from reuse of components/subsystems. Failure of a reused part whether as a consequence of the stresses incurred in the previous usage or from an error in inspection and/or refurbishment shall not constitute justification for classifying as a "no test". Such tests shall be classified as "failures".

3.1.3 Contractor/government furnished equipment requirements. Requirements for both contractor furnished equipments and government furnished equipments necessary for conducting and/or supporting component qualification and system service release testing are delineated in AR 72.

3.2 Test vehicle for performance demonstrations. The service release test vehicle, including those for testing cockpit capsule escape systems, shall conform to the requirements of MIL-A-81973. Throughout the designing and construction of the test vehicle, the contractor shall maintain an active liaison with the test activity to ensure compatibility of the test vehicle with the test track. System design verification tests (DVT) may be conducted using available test vehicles acceptable to the Government procuring activity.

3.3 Test facilities.

3.3.1 Apparatus.

3.3.1.1 Test equipment. The test equipment employed in conducting escape systems conditionings and tests shall be capable of (a) producing within the prescribed tolerances the specified conditions for the conditioning(s) and/or test(s) in which the equipment is used and/or (b) performing satisfactorily under the conditions specified for the conditioning(s) and/or test(s) in which the equipment is employed.

3.3.1.2 Test instrumentation. The test instrumentation used in conducting escape systems conditionings and tests shall be capable of (a) consistently and reliably measuring, transmitting and recording within the prescribed tolerances the data required for controlling and/or analyzing the conditioning(s) and/or test(s) in which the instrumentation is employed and (b) performing satisfactorily under the required conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.3.2 Test area. The area of the test locale(s) selected for test(s) shall ensure maximum safety of test personnel. Consideration shall be given to the protection of personnel and private property adjacent to the test area. The test area(s) shall be equipped with Government procuring activity approved optical measuring equipment to facilitate the accurate determination of the space position of escape system(s) and dummy(ies) during the test.

3.3.3 Test area access control. Control shall be exercised by the testing activity to preclude personnel not specifically authorized by the contractor test director, test facility test director and/or Government procuring activity test witness, from entering the test area during test preparations, testing, post-test data acquisition/test article recovery and inspection phases. Access to test articles shall be controlled strictly to ensure, particularly prior to post-test analyses, that articles are not disturbed and/or damaged. Failure to control access to test articles in the post-test recovery and analysis phase of a test may result in a Government classification of the affected test(s) as "no test" and requiring such tests to be reconducted at the expense of the activity responsible for the failure to maintain access control.

3.3.4 Frequency allocation. For all testing requiring the use of telemetry, whether at contractor, Government or independent test facilities, the contractor shall initiate in accordance with DD Form 1423, Contract Data Requirements List, appropriate frequency allocation applications DD Form 1494, Application for Frequency Allocation (Figure 1), directly with such Government agencies having approval cognizance, in sufficient time to ensure receipt of frequency assignments prior to the scheduled start of the affected tests. No tests requiring use of telemetry may be initiated prior to receipt of frequency assignments from the cognizant Government agency. The contractor shall coordinate all frequency allocation applications with the affected test activity and shall keep the Government procuring activity and/or its designee fully apprised of all actions affecting the submittal/processing of such applications.

3.4 Data acquisition. Measurements shall be made with instruments whose calibration accuracy and performance characteristics have been verified to the satisfaction of the Government procuring activity. Post-test evaluation of each test article shall be conducted thoroughly in accordance with MIL-STD-2067 requirements for failure/marginality of success analysis and the results of such examinations shall be provided to contractor design personnel for review and then appended with design section comments and recommendations to the test reports required under 3.6.3 and 3.6.4. Complete data and information describing the performance characteristics of the data acquisition devices shall be included in both the test plan and the test report.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.4.1 Electronic instrumentation. For all tests prescribed in the specification, electronic instrumentation shall be provided to sense, transmit and record, that data required by 3.4.4. In all cases the instrumentation shall provide coverage of, and record, crew member accelerations during the escape process, including pre-escape position, separation from the parent vehicle, entry into air stream, retardation and recovery system actuation, land and water impact, and other data as herein required. Additional electronic instrumentation may be specified and required at the option of the Government procuring activity. In addition, telemetry equipments (ground equipment, sled-mounted equipment, seat and/or dummy mounted equipment, etc.) shall comply with the requirements established by MIL-STD-442.

3.4.1.1 Structure-mounted instruments. The following data shall be taken from instruments mounted on the primary structure of the ejected escape system:

a. Escape path acceleration (Gz) of c.g. vs time.

b. Lateral acceleration (Gy) of c.g. vs time.

c. Longitudinal acceleration (Gx) of c.g. vs time.

d. Booster pressure vs time (may be sledborne instrumentation).

e. Rocket pressure.

f. Total drogue parachute loads vs. time (if applicable).

g. Event visual indicator(s).

The data shall be corrected for any offset between the mounting location and the center of gravity of the man-seat system.

3.4.1.2 Dummy-mounted instruments. The following data shall be recorded or transmitted by instruments mounted on the dummy:

a. Spinal acceleration (Gz) of chest vs time.

b. Lateral acceleration (Gy) of chest vs time.

c. Longitudinal acceleration (Gx) of chest vs time.

d. Pitch rate (Ry) vs time.

e. Roll rate (Rx) vs time.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

f. Yaw rate (Rz) vs time.

g. Right recovery parachute riser loads vs time.

h. Left recovery parachute riser loads vs time.

i. Event visual indicator(s).

3.4.1.3 Impulse noise level instrumentation. Selection of instruments for measuring and recording the impulse noise levels shall be selected on the basis of:

a. their response characteristics and accuracy

b. their ability to withstand without degradation of operation or fidelity, the environment imposed by escape system operation

c. their capability of being pre-test (especially post-installation) and post-test calibrated in a manner traceable to National Bureau of Standards

3.4.2 Photographic requirements. Photographic coverage requirements shall be delineated in either a Photographic Coverage Plan in accordance with DD Form 1423, Contract Data Requirements List, or as a section of the Test Plan (paragraph 3.5). Photographic coverage shall be provided for all tests prescribed herein to permit the measurement of position, velocity, acceleration, and attitude of the escape system. In addition, documentary and special event cameras shall be provided to record the complete test sequence, including the complete trajectory, and the behavior of pertinent and/or critical components or events of the escape system sequence during pre-ejection, ejection, separation, free-flight, and recovery subsystem operation phases. The behavior of pertinent and/or critical components or events shall include but need not be limited to:

(1) Behavior of and events associated with, interior equipments.

(2) Behavior of, and events associated with, ejection seats or capsules.

(3) Behavior of, and events associated with test dummies.

(4) Pre-ejection functions (if applicable).

(5) Cockpit canopy fracturing or jettisoning, or through-the-canopy ejection phenomena.

(6) Escape path clearance phenomena.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

(7) Ejection envelope clearance.

(8) Windblast effects upon dummies and their equipments, seat structures and canopies. Windblast effects upon windscreens and the effects of windscreens upon the windblast.

(9) Clearance between the dummy and seat combination and test vehicle (and thus aircraft) structure.

(10) Ejected system stability phenomena.

(11) Recovery parachute deployment and opening phenomena for each ejected system.

Still photographic coverage shall be provided as required herein. Additional photographic coverage may be specified and required by the Government procuring activity. Following each system test or series of component tests the contractor shall prepare a photographic coverage report bringing under one cover all significant pre- and post-test still photographic coverage and reference list of all motion picture coverage. Where feasible, all motion pictures coverage for a systems test or for a component test series should be combined on one reel, number and titled both on the film leader and on the reel to assure later ease of identification. The photographic coverage report may be presented as a section of the test report (section 3.6).

3.4.2.1 High-speed photographic coverage. Adequate high speed photographic coverage shall be used to furnish enough information for both detail and overall sequencing of trajectory analysis. Still photographic coverage shall include 70 mm format type cameras, or equivalent, and overall trajectory cameras such as Fairchild flight analyzer, or equivalent. (An acceptable equivalent to use of Fairchild flight analyzer would be the use of at least two fixed-position, pre-aimed motion picture cameras along with suitable markers in surveyed positions to provide a view of the entire trajectory. One camera to be located perpendicular to the track or flight path at the area of escape system initiation and separation from the test vehicle and one to be located in front of the test). The positions for high speed photographic coverage shall include a minimum of one front, one rear, one side, one cockpit mounted, and one mounted on sled nose. At least two cameras will provide a basis for making trajectory measurements.

3.4.2.2 Pre-test photographs. Still photographs shall be taken of the test article prior to, and subsequent to installation for each test. Particular emphasis shall be given to photographing the configuration/rigging of visible critical components/connections.

3.4.2.3 Post-test photographs. Following each test, still photographs shall be taken of any significant damage to components. Still photo-

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

graphs shall be taken of the undisturbed recovered condition of all components with particular emphasis for the recovered condition of those suspected to have malfunctioned or failed.

3.4.2.4 Other photographic coverage. Movie, video tape and still photographic coverage shall be available during the escape system fit and compatibility tests. Movie coverage shall commence upon the start of a formal demonstration and repeated to permit documentation of problems which have been noted. Still photographs illustrating particular problems which shall be taken for record purposes.

3.4.3 Timing correlation. Master timing systems shall be utililized in the instrumentation and photographic systems to aid data reduction and analysis. The master timing system shall provide a single timing source for both the photographic coverage and electronic data recording systems. Visual event initiation signals shall be incorporated upon both the test vehicle and the capsule or ejection seat escape system to provide means for correlating instrumentation records and photographic coverage.

3.4.4 Test data requirements. The following data shall be obtained during each ground, track and flight performance test;

a. Test site data

(1) Test site location

(2) Test site altitude

(3) The height above the ground, or reference datum line, of the dummy/seat combination c.g. at system initiation.

(4) Heading (Magnetic) of the test vehicle/aircraft during ejection.

b. Meteorological data

(1) Wind velocity and direction, peak gusts and directional variation, if applicable.

(2) Temperature

(3) Humidity

(4) Barometric pressure corrected to test site elevation.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

c. Three axis c.g. for each seat and seat/dummy combination.

d. Three dimensional mass-moment of inertia for each seat and seat/man combination.

e. Event times

(1) Canopy first motion or fracturing/severance or capsule severance system functioning.

(2) Canopy clear of escape path, if applicable

(3) Seat (capsule) first motion

(4) Rocket motor ignition

(5) Separation of ejection seat from aircraft

(6) Start of each stabilizer/decelerator action

(7) Completion of each stabilizer/decelerator action

(8) Separation of dummy from seat.

(9) Parachute pack opening

(10) Recovery parachute full riser line stretch

(11) Recovery parachute first full inflation

(12) Steady state descent (recovery)[1]

(13) Ground impact

f. Event heights

(1) Separation of dummy from seat

(2) Recovery parachute pack opening

(3) Recovery parachute full line stretch

(4) Recovery parachute first full inflation

(5) Steady state descent (recovery)[1]

1/ Steady state descent and recovery are defined as, "The point in the recovery phase at which the dummy achieves 30 feet per second (FPS) total velocity and 24 FPS vertical descent and never again exceeds these values."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

g. Graphs vs time of the following parameters.

(1) Seat/dummy (capsule) pitch

(2) Seat/dummy (capsule) roll

(3) Seat/dummy (capsule) yaw

(4) Seat/dummy (capsule) pitch rate

(5) Seat/dummy (capsule) rollrate

(6) Seat/dummy (capsule) yaw rate

(7) Seat/dummy (capsule) vertical displacement

(8) Seat/dummy (capsule) longitudinal displacement

(9) Seat/dummy (capsule) lateral displacement

(10) Seat/dummy (capsule) vertical velocity

(11) Seat/dummy (capsule) longitudinal velocity

(12) Seat/dummy (capsule) lateral velocity

(13) Seat/dummy (capsule) vertical acceleration

(14) Seat/dummy (capsule) longitudinal acceleration

(15) Seat/dummy (capsule) lateral acceleration

(16) Test vehicle acceleration[2]

(17) Test vehicle velocity[2]

(18) Test vehicle displacement

(19) Drogue riser loads

(20) Recovery parachute riser loads (right, left and total)

(21) Propulsion unit pressure history (through burnout)

(22) Clearance between seat/dummy and aircraft structure

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

(23) Total separation distance between each seat/dummy combination until surface impact

---

2/ The graphical display (as opposed to the data acquired) of test vehicle velocity and of test vehicle acceleration data shall include the period commencing 0.1 second prior to escape system initiation through no less than 0.1 second after the last dummy's heels clear the cockpit.

h. Maximum onset rate during the catapult stroke.

i. Vertical displacement vs longitudinal displacement for the canopy, seat and dummy (trajectory plot)

j. Vertical displacement vs lateral displacement for the canopy, seat and dummy (trajectory plot).

k. Dynamic pressure history on the dummy and equipment prior to, at the time of and immediately following ejection.

l. Pressure history in cockpit during high speed ejection tests.

m. Impulse noise level history in the vicinity of dummy's head (zero/zero tests).

n. Measurement and documentation of limb flailing.

o. IFF actuation signal.

p. Determination as to whether all escape system components functioned properly.

q. Report on visual inspection for permanent deformation of seat guide rails from bending, twisting, scoring and brinelling after each test.

r. Minimum clearance between escape system and aircraft structure.

Data acquisition shall commence at the start of the test and shall continue through all phases of the test until ground impact of the cockpit capsule of the last ejected dummy and equipment combination. To obtain these data, the contractor may utilize any commonly-accepted instrumentation methods complying with the requirements of this and subsidiary specifications, standards and publications. The contractor shall ensure that sufficient photographic coverage in accordance with 3.4.2 is provided both

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

inside and outside the test sled cockpit, to show cockpit capsule interior equipments, furnishings and dummies; cockpit capsule separation from sled; clearances between capsule and aircraft structure; capsule recovery parachute operation; or cockpit canopy jettisoning; ejection envelope clearance; windblast effect upon anthropomorphic dummies, seat structure, and wind screen; trajectories for the jettisoned cockpit, ejected dummy and equipment combination; clearance between dummies and aircraft structure and equipment; and personnel parachute opening for each ejected dummy. Any parachute damage which occurs shall be depicted upon a canopy damage chart, a sample of which is shown on Figures 2 and 3.

3.4.4.1 Test data analysis requirements. Data analysis for system and component tests shall include (a) full evaluation of all collected data for anomalous conditions and statistical distribution relative to specification limits, (b) full failure/marginality of success analysis of test articles and data in accordance with MIL-STD-2067, and (c) full analysis of test results, including data and failure/marginality of success analysis, to determine design and testing implications requiring design and/or test modifications. The contractor analyses shall include data verification steps in which the various test data shall be compared or otherwise checked for reasonableness. The proposed procedures shall be subject to the concurrence of the Government procuring activity.

3.5 Test Plan. Before starting any tests under this specification, the contractor in accordance with DD Form 1423, Contract Data Requirement List, shall prepare and submit to the Government procuring activity for approval, a comprehensive testing plan outlining all proposed test procedures and schedules and ensuring that the tests have been examined and designed to provide maximum benefit for all design evaluation efforts (i.e. reliability, maintainability, stress, marginality of success, etc). Flight tests shall not be performed prior to the satisfactory completion of tests of the propulsion subsystem, cartridges and cartridge actuated devices, and of the ground tests, unless otherwise specified by the Government procuring activity. The test plan should include test objectives and descriptions of each planned test, including equipment and facilities to be used, performance characteristics and accuracy of test instrumentation and recording devices to be used, and data to be recorded, and marginality of success plan. The test plan shall contain a section carefully delineating specific safety procedures to be followed by personnel conducting the test. Special attention shall be given to the procedures to be followed after an actual, or a suspected, hangfire or misfire of any explosive components and shall cover the waiting period before approaching the seat, method to be followed to safety the component, dismantling and handling procedures, and failure analysis procedures to be used. The test program shall include, but shall not be limited to the minimum demonstration and test regimen herein required. Purchases of costly test equipment, duplicating that available at Government facilities, shall be avoided by utilizing such facilities. In addition, the test plan for programs modified under the provisions of 3.8.4.3.4.1.1 shall include a data matrix conforming to the requirements of 3.6.1. This data matrix shall furnish the data, required herein, for all previous tests conducted on the escape system and utilized in obtaining a reduction in the test program.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

Data need not be provided for changes and tests occurring during the contractor's development program unless such tests are used in obtaining a reduction in the test program. The contractor shall submit the test program plan to the Government procuring activity at least sixty (60) days prior to scheduled commencement of the test program for comment or approval. Failure of the Government procuring activity to comment upon the test plan within thirty (30) days following the confirmed date of receipt shall constitute approval of the test plan. The contractor shall notify the Government procuring activity sufficiently in advance of the tests to permit a Government witness to attend.

3.5.1. Test plan change proposals. Subsequent to submittal of a test plan, any changes except those specified in writing by the Government agency designated as having approval authority, shall be submitted by the contractor in writing in Test Change Proposal (TCP) format (DD Form 1664) DID NO. DI-T-26391B).

3.5.2 Government facility test support document. As part of the test plan, the contractor shall prepare and submit to the Government procuring activity for approval a Government facility test support document detailing the specific test support required from, and the responsibilities of the Government facility. Preparation of the Government facility test support document shall be coordinated with each Government facility participating in the test program.

3.6 Reports of tests. Reports of tests will be specified on the contractor data requirements list. Numbers of reports, report distribution and submittal schedule required will be specified by the Government procuring activity. Copies of instrumentation and photography records shall be submitted with the reports. Motion picture films shall be furnished on reels suitable for use on projectors. Reports shall be prepared in conformance to MIL-STD-831, except as herein modified.

3.6.1 Test data presentation. To permit ready comparison of results obtained from the individual tests, the contractor shall prepare a matrix format for recording all test data obtained during each test program. The matrix shall indicate in the left-hand column the type of data recorded in each row. The first row shall include the test number and date. The second row shall include the acceptability classification of each run determined in accordance with the criteria specified herein. Data collected and presented shall include weather conditions observed at the start of the test (wind shall be described as a vector related to the test vehicle direction). The contractor shall include, in the matrix only, maximum values for time-history data. Any equipment failures shall be noted and an explanation of the failure appended. In

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

addition, for each Table II, III, IV and/or V test, the contractor shall prepare a standard escape trajectory plot in the Figure 1 format, using the scales shown in Figure 1 and noting thereon all major events, including but not limited to the following as applicable:

a. Canopy first motion or canopy severance.

b. Canopy clear of escape path.

c. Seat or capsule first motion.

d. Rocket ignition.

e. Start of each stabilizer/deceleration action.

f. Rocket burnout.

g. Completion of each stabilizer/decelerator action.

h. Man-seat separation.

i. Parachute pack opening.

j. Parachute full line stretch.

k. Parachute first inflation.

l. Achievement of steady state descent.

For Table II and III tests, the contractor shall prepare and include in the reports required by 3.6.3 and 3.6.5 a composite standard escape trajectory plot on which are shown the trajectories of all of the tests.

3.6.2 Preliminary report of test results (flash reports) Within 48 hours following the completion of a test of the system or a major component or subsystem thereof, the contractor shall prepare and submit to the Government procuring activity and its designated participating field activities a preliminary brief summary of the test and test results. The summary should include the test number and date, test article nomenclature, part number and serial number, test site location and testing activity identification, and a brief description of the test article configuration, the test conditions, test results, test article post-test condition, and any system/ component failures/malfunctions known or suspected to have occurred. The summary shall present in tabular form such preliminary data as can be obtained from instrumentation records/traces.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

3.6.3 Individual test reports. Within 20 days following completion of a test, and certified receipt of the data by the contractor, the contractor shall prepare and submit a detailed report, in the format outlined in Table VII, including the standard escape trajectory plot for the test, the updated composite standard escape trajectory plots (Tables II and III only), parachute damage charts (Figures 2 and 3) for all recovery, stabilization or other system parachutes, dummy damage charts (Figure 5) personnel equipment/ garments damage charts (Figure 7) and the data matrix covering all tests completed up to and including the test being discussed, for the test to the Government procuring activity. Copies of all instrumentation records and photographic coverage shall be included in the report. Movie films of the test shall be numbered and listed in the report.

3.6.4 Design verification test (DVT) reports.

3.6.4.1 Individual test/test series reports. The contractor shall prepare and submit to the Government procuring activity brief reports for each system and major subassembly/subsystem DVT conducted and for each series of component DVT conducted. The reports shall provide, as a minimum, the following information:

a. Test date.

b. Test location.

c. Test engineer's name and business affiliation.

d. Test article nomenclature and part number.

e. Test objective.

f. Test plan number and date.

g. Test conditions.

h. Test article configuration.

i. Names and business affiliations of witnesses.

j. Instrumentation listing.

k. Special test equipment listing.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

l. Test data.

m. Post-test analysis of test article.

n. Results of failure/MOS investigation.

o. Trajectory (if applicable).

p. Test classification.

q. Planned changes to remaining test plans and/or test article configurations.

The reports may be in the contractor's normal in-house "format."

3.6.4.2 DVT summarization report. Upon completing satisfactorily the design verification test program for the system and its elements, the contractor shall prepare and submit to the Government procuring activity a DVT summarization report. The report shall (a) identify all individual tests of systems and major subassemblies/subsystems and test series conducted and the associated reports, (b) trace all configuration and configuration change test history, and (c) provide for each test and/or test series a test classification. In addition, the report shall present the contractor's assessment and recommendations concerning the system's readiness to commence service release testing (SRT).

3.6.5 Final test report. Upon completion of all demonstrations and tests required herein, the contractor shall prepare a final test report summarizing the results of all demonstrations and tests performed and containing the complete data matrix developed during the system performance tests and all trajectory plots. In addition, the final test report shall include a list of all reports concerning the demonstration/ test program and a list of all applicable movie films. The report shall conform to the requirements of MIL-STD-831 as herein modified. The final report, in the format outlined in Table VII, shall contain a computed analysis, based upon the data collected during the systems performance tests herein specified, comparing the tested escape system's performance to that performance required by the applicable escape system specification. The contractor shall plot the predicted performance, specification requirements and system's actual tested performance in accordance with Figures 8 through 11 formats and include the charts in the final report. The program used for computing system performance, as well as the program verification shall be included. Three copies of the report shall be submitted to the Government procuring activity for review and approval and for approval of the escape system for service release. Within sixty (60) days following approval of the escape system for for service release, the contractor shall provide a reproducible and five (5) copies of the corrected final test report to the Government procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

3.7 Acceptance of systems. Systems or components may be considered satisfactory only if test results are acceptable to the Government procuring activity based upon criteria set forth in this specification. Approval and release of the system or component for installation and service use can be granted only by the Government procuring activity and will be based upon the recommendations of the Government witness(es), upon Government procuring activity evaluation of the data submitted in the reports required by 3.6, and upon both analytical and test demonstration of system conformance to design requirements. Systems will not be approved and/or released for installation and service use until the demonstrations and tests required herein for systems, subsystems and components have been completed satisfactorily. Figure 15 Aircrew Automated Escape System Design Evaluation Program Event flow Sequence Diagram (page 74) is provided to illustrate the time and general interrelationships between the program major elements and phases.

3.8 Types of service release demonstrations and tests. Prior to commencing any portion of the system service release test (SRT) program, including service release and/or qualification testing of any elements thereof, the contractor shall demonstrate to, and document for (3.6.4.1 and 3.6.4.2) the Government procuring agency, through an approved design verification test (DVT) program that the system and its elements are capable of complying with the design and performance requirements of the applicable military specifications. Following the submission of, and Government procuring activity approval of, the DVT summarization report (3.6.4.2), the aircrew automated escape system and its components shall be subjected to a service release test (SRT) program approved by the Government procuring activity prior to the commencement of any of the SRT. The SRT program, as a minimum, shall consist of the following types of demonstrations and tests:

a. Component/subsystem service release/qualification tests.

(1) Cartridges.

(2) Cartridge actuated devices.

(3) Rocket motor/rocket catapult propulsion subsystem and/or devices.

(4) Signal transmission subsystems.

(5) Ballistically-powered shoulder harness take-up device.

(6) Personnel or main recovery parachute subsystem.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

(7) Retardation and stabilization subsystem.

(8) Airspeed sensing and escape system mode of operation control subsystem.

(9) Survival equipment containers.

(10) Escape path clearance and canopy removal subsystem.

(11) Escape system controls.

(12) Personnel restraint subsystem.

(13) Impact attenuation subsystems.

(14) Survival beacon actuation.

(15) Survival beacon radiation pattern.

(16) Upholstery, covers, cushions and fabrics.

b. Aircrew station fit and compatibility tests.

(1) Manual egress demonstrations.

(2) Capability to reach and actuate aircraft controls.

(3) Capability to reach and actuate escape system controls.

(4) Aircrew percentile range accommodation.

(5) Field of vision.

c. Environmental conditioning and windblast testing.

(1) Environmental conditioning.

(2) Windblast loads tests.

(3) Explosive decompression tests.

(4) Electromagnetic compatibility (EMC) tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

d. Design compliance test.

(1) Crash loads strength demonstrations.

(2) Seat/man restraint physiological acceptability demonstrations.

(3) Ground, Track and flight performance demonstrations.

(4) Crash condition escape demonstrations.

(5) Underwater performance demonstrations.

(6) Impulse noise tests.

(7) Cabin air contamination.

e. Maintainability and reliability testing.

(1) Installed maintenance-free life demonstrations

All tests shall be planned, instrumented and conducted in a manner assuring the maximum acquisition of data necessary for evaluating and documenting system and element reliability.

3.8.1. Component/subsystem service release/qualification tests. Component/subsystem service release/qualification tests shall be conducted on a production type equipment and shall be completed satisfactorily prior to commencement of system performance testing. Component/subsystem service release/qualification tests shall be conducted in accordance with applicable military specifications and standards (except that salt spray conditioning of seat adjustment actuators shall be conducted in accordance with Procedure II of MIL-E-5272 in lieu of Procedure D. Seat adjustment actuator endurance testing shall include at least 10,000 full stroke cycles and 40,000 cycles of one inch stroke each. Three full stroke cycles shall be followed by 12 short cycles). In the event a component has completed satisfactorily a test program equivilent to the below stipulated requirements, the contractor may propose and request approval of a reduced test program as constrained by the applicable governing military specifications and standards for the component. The request shall be fully supported with design, configuration and test data and data defining the impact upon systems effectiveness aspects of the component and system. Following completion of each component/ subsystem test, all test articles shall be inspected for damage or excessive wear. The contractor shall prepare for Government procuring activity approval, service release/qualification test plans and specifications for critical components/ subsystems for which military specifications or standards either do not exist or do not define adequately the test requirements. In preparing the specifications, the contractor shall be guided by the requirements and practices set forth in MIL-STD-961 and shall coordinate testing requirements with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

requirements of MIL-STD-2067 to insure achievement of the reliability/maintainability requirements and acquisition of required reliability/maintainability data. The test plans and programs shall contain criteria for judging the success or failure of the tested articles to meet the requirements.

3.8.1.1 Cartridges. Cartridges shall complete satisfactorily all tests required by MIL-D-21625 and, in addition, all tests of the applicable cartridge actuated device(s) shall be completed satisfactorily.

3.8.1.2 Cartridge actuated devices. Cartridge actuated devices shall complete satisfactorily all tests required by MIL-D-23615.

3.8.1.3 Rocket motor/rocket catapult propulsion subsystem and/or devices. The rocket motor or rocket catapult type propulsion subsystem and/or devices shall complete satisfactorily all tests required by MIL-A-85041.

3.8.1.4 Signal Transmission subsystem testing.

3.8.1.4.1 Ballistic signal transmission line testing. Ballistic signal transmission lines shall complete satisfactorily all tests required by MIL-D-81980.

3.8.1.5 Ballistically-powered shoulder harness take-up device. The ballistically-powered shoulder harness take-up device shall have completed satisfactorily the qualification test requirements of MIL-D-81514, including the requirements for specific application testing (Table IA of MIL-D-81514).

3.8.1.6 Recovery parachute subsystem. The recovery parachute subsystem shall complete satisfactorily all MIL-STD-858 tests, or MIL-STD-888 tests, respectively, as required.

3.8.1.7 Retardation and stabilization subsystem. Prior to testing retardation and/or stabilization subsystems, the contractor shall prepare and submit to the Government procuring activity for approval, a test program plan for demonstrating the adequacy and acceptability of the retardation and stabilization subsystem performance and other characteristics for use in an aircrew escape system. The retardation and stabilization subsystem shall complete satisfactorily all tests of the test program.

3.8.1.7.1 Aerodynamic decelerators/stabilizers. The contractor-prepared test program plan for aerodynamic decelerators/stabilizers shall include, but not be limited, to the following tests:

a. All applicable MIL-STD-858 tests.

b. Maximum dynamic pressure bomb drop tests.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

c. Maximum altitude bomb drop tests.

d. Packed life/repack frequency tests to determine maximum safe packed life.

3.8.1.7.1.1 Phased aerodynamic decelerators/stabilizers. In addition to the requirements of 3.8.1.7.1, the test program plan for phased aerodynamic decelerators/stabilizers shall include, but not be limited to, the following phasing characteristics (i.e.: reefing/disreefing, staged franging, staged break-away, staged drag area reduction/rebuilding, staged squidding/desquidding, etc.) tests to demonstrate the consistency/repeatability, acceptability, and reliability of the phase operation(s):

a. Upper and lower airspeed limits for each airspeed-induced phasing operation.

b. Upper and lower time limits for each time-induced phasing operation.

c. Drag forces throughout each phase and throughout each transition between phases.

d. Capability of decelerator/stabilizer to perform all tasks required of it in applicable phases (i.e.: stabilizing man-seat combination, deploying personnel parachute, etc.)

3.8.1.7.2 Rocket motor type decelerators/stabilizers. In addition to the requirements of 3.8.1.7, rocket motor type decelerators/stabilizers shall complete satisfactorily all MIL-A-85041 tests.

3.8.1.7.3 Retardation and stabilization subsystem ballistic elements. All cartridges and cartridge actuated devices employed in the retardation and stabilization subsystem shall complete satisfactorily the applicable tests of MIL-D-21625 and MIL-D-23615, respectively.

3.8.1.8 Escape parameters sensing and escape system mode of operation control subsystem. The contractor shall prepare and submit to the Government procuring activity for approval a test program for demonstrating satisfactorily performance of unsealed sensor and control subsystems after prolonged exposure to aircrew station environments. In addition, the test program shall include tests for demonstrating the sensitivity and reliability of the subsystem for controlling the escape system mode of operation for ensuring safe escape. The escape parameters sensing and escape system mode of operation control subsystem shall complete satisfactorily all tests of the test program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.1.9 Survival equipment containers. Rigid seat survival kit (RSSK) containers shall complete satisfactorily all tests required by MIL-S-81018 or MIL-S-81040, as applicable. For survival equipment containers not conforming to the requirements of MIL-S-81018 or MIL-S-81040, the contractor shall prepare and submit to the Government procuring activity for approval, a test plan demonstrating the acceptability of the proposed configuration for use in aircrew escape system. All tests of the test program shall be completed satisfactorily.

3.8.1.10 Escape path clearance and canopy removal subsystem. The contractor shall prepare and submit to the Government procuring activity a test program for demonstrating the acceptability and reliability of the operation of the escape path clearance and canopy/hatch removal subsystem. All ballistic elements shall complete satisfactorily all tests required by MIL-D-21625, and MIL-D-23615, as applicable, prior to commencement of subsystem service release tests. Canopy removal tests shall be conducted in accordance with the requirements of MIL-C-81590 and this specification. The escape path clearance and canopy removal subsystem shall complete satisfactorily all tests of the test program.

3.8.1.11 Escape system controls. The contractor shall prepare and submit to the Government procuring activity for approval a test program for ensuring the escape system controls meet (a) pull force requirements for extraction from their housing and for initiation of the escape sequence, (b) snatch and ultimate loads, and (c) cycle life requirements as specified by the applicable escape system specification. The tests shall include operation of the personnel restraint release handle. All escape system and seat controls shall complete satisfactorily all tests of the test program.

3.8.1.12 Personnel restraint. The contractor shall prepare and submit to the Government procuring activity for approval a test program for ensuring the personnel restraint subsystem complies with requirements of the applicable escape system specification. The impact attenuation subsystem shall complete satisfactorily all tests of the test program.

3.8.1.13 Upholstery, covers and cushions.

3.8.1.13.1 Flame resistance. Samples of all upholstery and cover material shall be tested in accordance with method 5906 of FED-STD-191. The burning rate shall not exceed 4 inches per minute. In addition, if the specimens do not support combustion after the ignition flame is applied for 15 seconds, or if the flame extinguishes itself and subsequent burning without a flame does not extend into the undamaged areas, the material shall be acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.2 Aircrew station fit and compatibility tests. The aircrew station fit and compatibility tests (constituting the In-Process Design Conformance Inspections, IDCI) shall be conducted using the engineering proofing kit installed in either an actual aircraft or mockup acceptable to the Government procuring activity. These tests shall be completed satisfactorily prior to the commencement of system performance tests. The contractor shall supply for each fit and compatibility test a sufficient quantity of "In-process Design Conformance Inspection" forms in the Figure 12 format for use by reviewers in recording deficiencies.

3.8.2.1 Mock-up role in aircraft fit and compatibility tests. The escape system mock-up shall be installed in the aircraft, or an aircraft cockpit/ aircrew station mock-up acceptable to the Government procuring activity, for a preliminary evaluation of escape system fit in, and compatibility with, the aircraft and for preliminary evaluation of aircrew accommodation and mission task performance.

3.8.2.2 Engineering proofing article role in aircraft fit and compatibility tests. The escape system engineering proofing article shall be installed in the aircraft, or an aircraft cockpit/aircrew-station mock-up acceptable to the Government procuring activity for evaluation of escape system fit in, and compatibility with, the aircraft and for evaluation of aircrew accommodation and mission, task performance.

3.8.2.3 Production proofing article role in aircraft fit and compatibility tests. The escape system production proofing article shall be installed in the aircraft for final verification of escape system fit in, and compatibility with, the aircraft and of aircrew accommodation.

3.8.2.4 Manual egress demonstrations. The contractor, utilizing both 3rd and 98th percentile aircrew subjects equipped with personnel parachutes if applicable, and personnel survival equipment, as applicable, shall demonstrate to the satisfaction of the Government procuring activity aircrew manual egress/escape capability. Should aircrew escape be impaired when currently approved egress procedures are utilized, the contractor, in order of preference, shall (a) redesign the systems involved to remove the cause of egress impairment, or (b) develop and demonstrate new egress procedures to overcome the impairment. If new manual egress procedures are approved by the Government procuring activity, the contractor shall prepare changes for flight manuals describing and illustrating the new procedures and shall ensure that all applicable training aids, being prepared under contract, reflect the changes.

3.8.2.5 Demonstration of aircrew capability to reach and actuate aircraft controls. For each aircrew station, the contractor shall demonstrate compliance with the requirements of 3.3.1(b) of MIL-S-18471 or MIL-A-23121, as applicable.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.2.6 Demonstration of aircrew capability to reach and actuate escape system controls. For each aircrew station, the contractor shall demonstrate compliance with the requirements of 3.3.1 (a) and (c) of MIL-S-18471 or MIL-A-23121, as applicable.

3.8.2.7 Demonstration of aircrewman percentile range accommodation. For each aircrew station, the contractor shall demonstrate compliance with the requirements of 3.3.1(a) of MIL-S-18471 or MIL-A-23121, as applicable, concerning aircrewman accommodation.

3.8.2.8 Field of vision. The field of vision plots for each aircrew station shall be verified as part of the compatibility demonstrations.

3.8.3 Environmental conditioning and windblast testing.

3.8.3.1 Environmental test regimen. The environmental testing of the escape system shall be conducted on at least one complete, inert system Humidity, vibration and salt spray tests shall be conducted in accordance with MIL-STD-810. The tests shall be performed in the order recommended under the heading, "Mechanical Systems", in Table II of MIL-E-5272.

a. High temperature tests, MIL-STD-810, Method 501, Procedure II

b. Low temperature tests, MIL-STD-810, Method 502, Procedure I

c. Humidity tests, MIL-E-5272, Procedure I.

d. Temperature-altitude tests, MIL-STD-810, Method 504, Procedure II

e. Vibration conditioning (Operational Vibration), MIL-E-5272, Procedure XII (both with and without occupant).

f. Fungus resistance tests, MIL-STD-810, Method 508, Procedure I

g. Sunshine tests, MIL-STD-810, Method 505, Procedure I

h. Rain tests, MIL-STD-810, Method 510, Procedure I

i. Sand and dust tests, MIL-STD-810, Method 510, Procedure I

j. Shock tests, MIL-STD-810, Method 516, Procedure I

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

k. Acceleration tests, MIL-STD-810, Method 513, Procedures I and II

l. Salt spray tests, MIL-E-5272, Procedure II (Sulphur dioxide gas shall be injected into the salt spray chamger at six hour intervals. The gas shall be fed in for a period of sixty minutes at a gas flow rate of 50 cc per hour per cubic foot of test chamger volume.

Following each test, the escape system shall be inspected and the condition of the system and its components noted. In the event corrosion is noted, color still photographs shall be taken and sketches shall be made to record its extent. Serious corrosion or other evidence of failure of the escape system to withstand environmental conditioning satisfactorily shall constitute evidence of nonconformance with the design requirements of MIL-S-18471 or MIL-A-23121, as applicable, and may result in redesign and retesting. In the event a component and/or finish specification requires environmental testing, such testing shall have been completed prior to the start of the system environmental testing herein specified.

3.8.3.2 Windblast loads tests. Before commencing tests required herein, the fully-equipped ejection seat with a dummy properly restrained for ejection shall be exposed twice to windblast of 600 ±10% KEAS (or maximum aircraft airspeed ±10%, if less than 600 KEAS) for a minimum of 0.3 second per exposure in each of the following attitudes:

a. Emerging into airstream during ejection (fully exposed, in line with guide rails).

b. 45° rotation to right.

c. 90° rotation to right.

d. 45° rotation to left.

e. 90° rotation to left.

f. 45° pitch forward.

g. 90° pitch forward.

h. 45° pitch aft.

i. 90° pitch aft.

Photographic coverage of each windblast exposure test shall include a minimum of two (2) high-speed motion picture cameras (2,000 fps or greater), located to provide maximum coverage of exposed portions of the system. All failures shall be evaluated to ascertain whether they might cause injury to the crewman, damage to survival equipment and/or protective garments, and/or degrade escape system performance.

3.8.3.3 Explosive decompression tests. The entire production configuration escape system shall be subjected to an explosive decompression from 8,000 feet equivalent altitude to the pressure equivalent of the maximum cruise altitude of the aircraft. Decompression time shall be 100 milliseconds or less. After this exposure, operation of the escape system controls and seat adjustment actuator shall be tested and proper operation of all

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

personal services verified. All escape system components shall be inspected for damage that could degrade aircraft and/or escape system performance or cause injury to crewmen or damage to their personal protective garments or survival equipments.

3.8.3.4 Electromagnetic compatibility tests. Electromagnetic compatibility tests shall be conducted in accordance with MIL-STD-461, MIL-STD-462 and AR 46 to verify escape system compatibility with the aircraft electromagnetic conducted and radiation interference environment and requirements. At least one sample of each electrically different seat shall be tested in each of the modes listed below. If all seats are electrically identical, only the pilot's seat need be tested.

a.	Steady State	Seat motor continuously operating
b.	Transient	Start/stop motor operation
c.	Transient	Power applied to motors until limit is reached.

3.8.4 Design compliance tests. Design compliance tests shall be conducted to ascertain compliance of the escape system and/or seats with specified design requirements crash loads strength and escape performance requirements. In addition, the tests shall be conducted to verify that: (1) the cockpit capsule will separate from the aircraft, clear aircraft structure and function correctly to provide safe escape; or (2) the canopy will jettison cleanly from the cockpit, the ejection seat and occupant will separate from the aircraft and clear aircraft structure satisfactorily, and the escape system will function correctly to provide safe escape, throughout the required escape envelope.

3.8.4.1 Crash loads strength demonstration. The contractor shall demonstrate compliance with the crash loads strength requirements of the applicable escape system specification by the successful application of the required dynamic loads to two complete ejection seat systems, including mounting provisions, or to the furnishings and escape system components located in cockpit capsules with fully equipped 98th percentile dummies installed. In the event mounting provisions for individual type aircrew escape systems vary among the aircrew stations, each variation will require two successful tests of the complete systems involved unless the Government procuring activity, on the basis of design review, waives the requirements.

3.8.4.2 Seat/Man restraint physiological acceptability demonstrations. Seat/man restraint physiological acceptability demonstrations, as defined in Table I are conducted to demonstrate the physiological acceptability

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

of new escape systems and/or changes that affect man restraint, position and/or loading in existing escape systems. Escape system changes requiring verification of the physiological effect upon the escape system occupant(s) include, but need not be limited to, the following:

a. Change in boost catapult thrust characteristics.

b. Change in sustainer motor thrust vector.

c. Use of new restraint subsystem.

d. Change in seat platform shape, mounting, cushion and/or angular relationship to thrustline and/or back support.

e. Change in back support/lumbar support angular relationship to thrustline and/or seat platform.

f. Change in location of fully-equipped seat occupant's head relative to his back tangent line.

g. Changes in limb restraint subsystem.

h. Changes in man restraint subsystem seat attachment points.

These tests will be conducted with the escape system seat mounted on an ejection tower carriage so as to reproduce the escape acceleration loads vector and onset rates upon the seat occupant.

WARNING:

To ensure subject safety, the seat configuration employed in these demonstrations shall have completed successfully the crash testing and the actual seal shall have been subjected successfully to approved production proof load and structural inspections.

3.8.4.3 Ground, track and flight performance demonstrations. The contractor shall demonstrate, as provided herein, compliance with the escape capability performance requirements of the applicable escape system specification.

3.8.4.3.1 Prerequisite for testing. The contractor shall not commence the ground, track and/or flight performance demonstrations until the Government procuring activity has approved the contractor's test plan and the demonstrations and tests required under 3.8.1, 3.8.2, 3.8.3, 3.8.4.1, and 3.8.4 have been completed successfully.

3.8.4.3.2 Definitions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.4.3.2.1 System zero speed and low speed operational tests. These tests are conducted to demonstrate the performance capability, proper integration within the aircrew station to ensure adequate escape envelope, proper operation, structural integrity, and functioning of the complete escape system and interrelated vehicle components and systems from the initiation of the escape sequence to ground recovery of the dummy and, if applicable, the escape system. These tests shall be conducted from the aircraft, test sled or from a suitable fixture which can be mounted on a vehicle for the low speed operational tests. The fixture shall simulate the applicable aircraft and cockpit structure in a manner acceptable to the Government procuring activity.

3.8.4.3.2.2 Track tests. Track tests are those tests conducted in accordance with Tables II and III to demonstrate the proper functioning of the ejection, separation, stabilization, and recovery subsystems and components at velocities representing the complete dynamic pressure range of the aircraft or vehicle. The tests may be conducted from a jet-propelled sled or from a rocket-propelled sled representing the aerodynamic configuration of the aircraft structure adjacent to the escape system and containing the escape system and associated structure. The contractor shall consider economic factors in choosing the test vehicle propulsion mode.

3.8.4.3.2.3 Flight tests. Flight tests as defined in Table IV, are conducted to demonstrate and functionally verify ejection, separation, stability, onboard sensing and control subsystems, and recovery subsystems and components at velocities, sink rates, altitudes, and attitudes (including adverse attitudes) representing the performance envelope of the escape system. The tests shall be conducted from an aircraft, upon approval by the procuring activity.

3.8.4.3.2.4 Crash condition escape tests. Crash condition escape tests, as defined in Table V, are conducted to demonstrate and functionally verify that all escape system components will operate in a manner ensuring safe escape and recovery of crew members under crash conditions requiring automated crew escape. Such escapes normally occur under varying moderate deceleration loads and may be necessitated by fire, impending collision, etc. The tests shall be conducted from a test fixture or test vehicle capable of withstanding the deceleration loads and approved by the Government procuring activity.

3.8.4.3.2.5 Underwater escape capability tests. Underwater escape tests as defined in Table VI are conducted to evaluate and determine procedures/techniques to be used to effect underwater escape.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

3.8.4.3.2.5.1 Canopy implosion tests. To ensure manual egress test subject safety, prior to the ejection tests, the cockpit section, with canopy closed, will be lowered in the water to determine depth and pressure differential where implosion occurs to ensure compliance with MIL-C-81590, MIL-S-18471 and MIL-A-23121, as applicable.

CAUTION

Before commencing tests under 3.8.4.3.2.5.3, implosion test (3.8.4.3.2.5.2) must be completed and safety precautions established and approved.

3.8.4.3.2.5.2 Manual underwater escape capability tests. Manual underwater egress tests will be conducted from a submerged cockpit section or cockpit mock-up section rigged such that it can be rotated to, and supported in, taildown and inverted positions. The cockpit section shall be configured with controls such as (stick and rudder pedals), consoles and any other equipment that might affect egress, as well as the complete functional escape system with the exception of the main propulsion pyrotechnics and/or rocket. Test subjects for the manual egress tests shall be trained SCUBA divers. The tests shall be conducted in a facility possessing controlled water conditions. Qualified subjects, approximating 3rd and 98th percentile in stature shall wear complete aircrewman's personal flight gear in each of the various configurations of flight and environmental protective garments authorized for use in the aircraft. The egresses will be made in each of the most adverse seat adjustment positions from each of the four cockpit attitudes (defined in Table VI) to ascertain the optimal means for effecting manual underwater escape and to evaluate problems such as snagging, proper operation of the equipment in the water environment, etc. In multiplace aircraft, egress tests will be made from each aircrew station. Safety precautions as established by the testing facility will be followed meticuously and all equipment will be new, checked and certified prior to test by the contractor. Photographic coverage and subject debriefing will be used in evaluating the egress tests.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:54Z
Check the source to verify that this is the currenversion before use.

3.8.4.3.2.6 Thermal/radiation closure tests. Thermal/radiation closure tests are conducted to evaluate safety of escape from aircraft so equipped.

3.8.4.3.3 Procedures.

3.8.4.3.3.1 Escape path clearance tests. Escape path clearance tests shall be performed in conjunction with ejection seat tests specified in Tables II and III. Escape path clearance shall be accomplished by direct operation of the firing control unless other means have been approved by the Government procuring activity prior to commencement of testing.

3.8.4.3.3.2 Through-the-canopy, canopy removal by seat and through thermal/radiation closure tests.

3.8.4.3.3.2.1 Through-the-canopy and canopy removal by seat tests. Those aircraft having the capability of ejecting through the canopy or having the seat forcibly remove the canopy, either as a primary or as a secondary means of escape, shall have at least two tests conducted in this mode of escape.

3.8.4.3.3.2.2 Through thermal/radiation closure tests. Those aircraft equipped with and having the potential for ejecting through, thermal/ radiation closures, either as a primary or as a secondary means of escape, shall have at least two tests conducted with the thermal/radiation closures in the closed position (i.e: protective versus stowed position) and semistowed position which has been determined to pose the most critical danger to safe escape. These may be combined with the tests conducted in accordance with 3.8.4.3.3.2.1.

3.8.4.3.3.2.3 Special data acquisition/analyses. For the tests conducted in accordance with 3.8.4.3.3.2.1 and 3.8.4.3.3.2.2 special attention shall be given to the selection of instrumentation, including photographic, and data reduction and analyses techniques to measure, record and evaluate head impact loads, peak accelerations, acceleration onset rates, jolts, velocity change, and other physiological and system mechanical performance parameters useful in determining safeness of the mode of escape.

3.8.4.3.3.3 Ejected weight. The weight of the ejected portion of the escape system shall be established by the contractor and verified by the testing activity. The ejected weight shall be representative of that planned

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

for use under service conditions and shall include the weight of the items of personnel equipment specified for use with the system, crew member weight (3 through 98 percentile) and the seat weight. The three axis center of gravity (cg) of the ejected mass shall be located to simulate service ejection conditions. For systems employing lateral divergence or other lateral acceleration inducing techniques, the contractor shall consider and establish whole body and internal organ displacements of humans, variations between human lateral centers of gravity, and potential variations in body positioning within the system. The location of the rocket thrust line with respect to the ejected mass-cg distribution pattern is particularly critical for rocket propelled escape systems and tests shall be performed, therefore, at extremes of cg-thrustline misalignment.

3.8.4.3.3.3.1 Ballasting. Ballasting to achieve a specific required ejected system center of gravity must be performed with care to avoid inducing significant changes in the moments of inertia of the ejected system and, in the case of non-capsular systems, the centers of gravity and moments of inertia of the ejection seat, the dummy, the dummy's limbs, and the dummy's head. Failure to exercise care in locating and sizing ballast can induce non-representative motions and loads capable of masking the true system performance. Accordingly, the use of ballasting shall be kept to a minimum and, when used, the contractor shall record in the test report the pre- and post-ballasted weights, centers of gravity, and moments of inertia for the ejected system and each separable component thereof, including dummy limbs, head and torso. Any ballasting in excess of 5 percent of the ejected system weight or of the weight of the element to which attached shall require presentation of all pertinent information to the Government procuring activity test witness for review prior to the test. A study correlating human subject and dummy seated position static and dynamic centers of gravity shall be employed to ascertain the need for and the appropriate amount and location(s) of ballasting.

3.8.4.3.3.4 Escape system initiation. The means employed during tests to initiate the escape system shall be subject to review and approval by the Government procuring activity.

3.8.4.3.3.5 Configuration. An instrumented anthropomorphic dummy shall be secured to the seat by the personnel restraint system provided the crewman. The seat shall be adjusted so that the position of the ejected

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

mass center of gravity has the most critical effect upon safe escape. The dummy center of gravity shall closely approximate the center of gravity of the appropriate percentile crewman subjected to acceleration loadings.

3.8.4.3.3.6 Ejection seat recovery equipment. If a seat recovery parachute can be mounted without significant effect upon the escape system or seat/man combination center of gravity, it shall be utilized where possible to reduce landing impact damage to the test article and permit post-test examination of relatively undamaged escape system components. Use of a recovery parachute shall require Government procuring activity approval. Seat recovery lanyards attached either to the dummy or to the personnel parachute subsystem shall not be employed.

3.8.4.3.3.7 Instrumentation (recording and evaluating data). The following criteria shall be used in assessing test data necessary for determining the capability of the escape system or its components to meet prescribed physiological acceleration tolerance.

(a) The peak acceleration, Gx, Gy and Gz, (1) shall be measured in the portion of the acceleration curve achieving the greatest magnitude and (2) shall be that highest level sustained for at least 20 milliseconds (Figure 13).

(b) The rate of acceleration onset, Gx, $\dot{G}y$ and $\dot{G}z$, (1) shall be measured in the initial portion of the acceleration curve and (2) shall be that rate defined by the greatest change in acceleration occurring in any 30 millisecond interval, i.e.

$$\dot{G} = \frac{\Delta a \text{ max}}{0.030 \text{ sec.}}$$

(Figure 14).

(c) Pitch (R_y), roll (R_x) and yaw (R_z) rates shall be the highest rates of attitude change averaged over an interval of 30 milliseconds and shall be expressed in RPM (revolutions per minute).

(d) Identification of negative and positive values for accelerations (G), rates of acceleration onset (G) and rotations (R) shall be in accordance

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

with the standard right hand rules in which the thumb, fore finger and middle finger of the right hand are extended ninety degrees to one another (thumb pointed vertical, upwards) indicate the positive directions, for G and G. For R, the righthand thumb is pointed in the positive direction of the axis about which the rotation occurs and the finger curled in towards the palm. Direction of the fingers then indicates positive direction of rotation about that axis.

(e) Parachute riser loads shall be taken from that portion of the topmost peak of the force time curve lasting 20 milliseconds.

(f) Maximum total parachute riser loads shall be taken from that portion of the topmost peak of the resultant (combined) force time curve lasting 20 milliseconds.
(NOTE: The maximum total riser loads thus may be equal to, or less than, the sum of the right and left riser loads.)

3.8.4.3.3.7.1 Hard line data transmission. Hard line (or umbilical) data transmission shall not be employed for acquisition of data from the ejectable portion, including dummy, of the escape system.

3.8.4.3.4 Performance tests. Escape system performance tests shall be conducted as herein specified, unless otherwise directed by the Government procuring activity. The escape system performance test program shall consist of, but need not be limited to, the tests established by 3.8.4.4, 3.8.4.5, 3.8.4.6, 3.8.4.7, and 3.8.4.8 and Tables II, III, IV and V herein, as applicable and as modified by the Government procuring activity under the provisions of 3.8.4.3.4.2.1. and reliability/marginality of success/failure analyses required by MIL-STD-2067. The escape system performance test program shall be developed in accordance with the following:

a. Non-cockpit capsule individual type escape system.

(1) Single crew station. Single crew station escape system performance testing shall conform to the requirements of Tables II, IV and V and paragraphs 3.8.4.4, 3.8.4.5, 3.8.4.6, 3.8.4.7, and 3.8.4.8 as a minimum.

(2) Multi-crew station. Multi-crew station escape system performance testing shall conform to the requirements of Table II, IV and paragraphs 3.8.4.4, 3.8.4.5, 3.8.4.6, 3.8.4.7, and 3.8.4.8 as a minimum.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

b. Cockpit capsule type escape system. Performance testing for both single crew and multi-crew cockpit capsule type escape systems shall conform to the requirements of Table III and V and paragraphs 3.8.4.4, 3.8.4.5, 3.8.4.6, 3.8.4.7 and 3.8.4.8 as a minimum.

Escape system zero-zero, 50 kt. and crash condition escape performance tests shall be conducted from either a test fixture or a test vehicle duplicating the crew station(s) and, if applicable, cockpit capsule-aircraft separation elements. Both 3rd and 98th percentile anthropomorphic dummies shall be ejected in seats positioned to simulate the most adverse escape system/man center of gravity. In the 50 kt. tests the escape system attitude shall be to simulate the conditions obtained when the aircraft is (1) 20 degrees nose down (simulating loss of a nose wheel) and (2) 30 degrees bank (simulating loss of one main gear and/or sliding off a flight deck). Track tests shall be conducted from a test vehicle which duplicates the crew station(s) and, if applicable, cockpit capsule-aircraft separation elements. The test vehicle shall be capable of being propelled along a ground track at the required speeds. Flight tests shall be conducted from a test aircraft specifically equipped for conducting inflight escape system tests. In the event that escape system configuration varies between crew stations, at least one configuration shall be subjected successfully to all of the Table IV tests and each of the other configurations shall be subjected successfully at least to two different tests each from Table IV.

3.8.4.3.4.1 Performance test program. To be released for service use, an escape system shall complete satisfactorily the minimal programs specified in Tables II, III, IV and V, as applicable, in accordance with the performance criteria specified in the applicable escape system design specification. In addition, as provided herein, the Government procuring activity may require the satisfactory completion of the evaluation programs of Tables I and VI.

3.8.4.3.4.1.1 Test program modifications. In the event the escape system previously has completed satisfactorily a test program and has been released for installation and service use in other aircraft, the contractor may propose a reduced test program. The contractor , by presenting supporting design, configuration and configuration control, and test data, may propose a one-for-one reduction in the Tables II, III, and IV test programs in those instances where qualification type tests (i.e. with test article configuration and configuration control, testing data collection and photographic coverage complying with the requirements of this specification) have been conducted on the same configuration ejection seat proposed. The minimal acceptable reduced test program shall demonstrate achievement of AAES reliability conforming to MIL-STD-2067 requirements. Approval of the proposed reduced test programs shall be at the discretion of the Government procuring activity. In computing AAES reliability the contractor in accordance with the provisions of MIL-STD-2067 may utilize the results of those tests proposed to and accepted by the Government procuring activity as a basis for reducing the test program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.4.3.4.1.2 Test program schedule. The test program schedule shall permit adequate time for analysis of escape system test data during the testing. As a minimum, test data shall be analyzed and the results presented for Government procuring activity review before the conduct of the third following test prior to proceeding with that test. If the testing is conducted at and by a Government test facility, the contractor shall advise the Government procuring agency and/or its designee concerning any and all instances in which delivery of test data is delayed, or is likely to be delayed, beyond the scheduled data delivery date.

3.8.4.3.4.2 Purpose of tests. The purpose of the performance demonstrations shall be to:

a. Provide data for assessing system reliability conformance to requirements of MIL-STD-2067.

b. Detect and identify for management attention any actual and/or incipient problems which may degrade system performance and/or safety (through MOS and failure analyses conducted in accordance with MIL-STD-2067).

c. Furnish sufficient information to verify escape system performance compliance with the requirements of MIL-S-18471 or MIL-A-23121, as applicable. (Critical areas shown to exist by computer simulation analyses may necessitate further tests in order to correct these deficiencies.)

d. Obtain data for preparing the NATOPS manual presentations required by MIL-A-81573.

3.8.4.4 Underwater performance demonstrations. The contractor shall prepare and submit to the Government procuring activity a program for demonstrating compliance with the underwater escape system/components requirements of the applicable escape system/components specification. The escape system/components shall complete satisfactorily all tests of the program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

3.8.4.5 Impulse noise levels. During the zero-zero performance demonstrations, impulse noise level measurements shall be made of the escape system operation (1) from system initiation through heel clearance with cockpit (MIL-S-18471) or (2) from initiation until completion of all system functions (MIL-A-23121) to demonstrate compliance with 3.2.1 of MIL-S-18471 or MIL-A-23121, as applicable. The contractor shall submit to the Government procuring activity for approval a description of the noise measuring equipments proposed for use during the test. Instrumentation shall conform to the requirements of 3.4.1.3, herein.

3.8.4.6 Cabin air contamination tests. For cockpit capsule and encapsulated ejection seat type escape systems, the contractor shall prepare and submit to the Government procuring activity for approval a test program for collecting and analyzing cabin air during and immediately following escape system functioning. Prior to conducting the zero-zero tests, the program shall describe the methods and equipments to be used in the collection and analysis of the cabin air samples. During the systems performance testing herein specified, cabin air samples shall be collected and shall be analyzed in accordance with the approved test program. Contaminant levels shall not exceed those permitted by MIL-A-23121.

3.8.4.7 Flotation tests. The contractor shall prepare and submit to the Government procuring activity for approval an escape system flotation subsystem test program plan if applicable. Non-cockpit capsule type escape systems providing crew flotation automatically shall be subject to tests designed to verify the acceptability of the flotation system operation and flotation characteristics. Cockpit capsule escape system testing shall be designed to prove compliance with the flotation requirements of MIL-A-23121.

3.8.4.8 Field maintainability verification. The contractor shall furnish to a Government activity designated by the Government procuring activity a prototype escape system thirty (30) days prior to, or thirty (30) days following presentation of the escape system engineering proofing article and a production, inert escape system within thirty (30) days following presentation of the production proofing article. Each escape system shall be subjected to a field maintainability verification conducted in accordance with maintainability demonstration test plan prepared in accordance with MIL-STD-2067 and approved by the Government procuring activity by the designated Government activity to ensure system/component maintainability, to develop information for evaluating contractor recommended maintenance procedures and practices for evaluating GSE and SSE (ground support equipment and special support equipment) suitability for performing the required maintenance tasks in a fleet maintenance environment, and to evaluate susceptibility of escape system components to damage during cockpit aircraft maintenance or equipping. All ballistic components of the prototype escape system shall be inert, however, all mechanical firing mechanisms shall be in working order.

3.8.4.8.1 Field maintainability evaluation article I (FMA I) function. The function of the FMA I is to serve as a means for assessing AAES design capability of meeting all specified maintainability requirements. Delivery of the FMA I

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

should be accomplished in a timely manner, permitting completion of the review and the subsequent design deficiency correction incorporation in service release test articles and production articles.

3.8.4.8.2 Field maintainability article II (FMA II) function. The FMA II shall be a production article and shall be used to evaluate contractual compliance.

3.8.5 Maintainability and reliability testing. Maintainability and reliability testing shall be conducted in accordance with the requirements established in MIL-STD-2067.

3.8.5.1 Installed maintenance-free life demonstrations. The contractor shall identify all age-sensitive and potentially age-sensitive elements, the potential effects of each environmental factor specified by the system specification, and the design approaches employed to minimize and/or retard environmental degradation of the system elements. Using these data the contractor shall prepare and submit to the Government procuring activity or its designee a plan for demonstrating achievement of the installed maintenance-free life requirements of MIL-STD-2067. Among approaches the contractor should consider for inclusion in the demonstration plan are:

- component environmental conditioning.
- system environmental conditioning.
- longterm (real time) environmental conditionings
- fleet parts return surveillance.

3.9 Comparison of tested escape system's performance against performance requirements. Utilizing data obtained during the systems performance tests herein specified, the contractor shall perform a computational analysis of the tested escape system's performance to verify compliance with the performance requirements of MIL-S-18471 or MIL-A-23121, as applicable. Test data utilized for formulation of equation, or computer programs, shall not be used for verifying the accuracy of the equation and/or program. In addition, following completion of the test program, the computer analyses shall be used to formulate performance charts and data conforming to the MIL-A-81573 formats.

3.10 Prerequisites for commencing service release test (SRT) program. Prior to commencing the service release test (SRT) program defined in accordance with 3.8, the contractor shall have completed to the satisfaction of the Government procuring activity (1) the mock-up article (MUA) and engineering proofing article (EPA) in-process design compatibility inspections required by MIL-S-18471 or MIL-A-23121, respectively, (2) a component/ system design verification test (DVT) program developed by the contractor and approved by the Government procuring activity to demonstrate acceptable component/system functioning, performance, integration, and interrelationship, (3) full definitization of an acceptable baseline configuration for service release testing in accordance with MIL-S-18471 or MIL-A-23121, as applicable,

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

and (4) completion acceptable to the Government procuring activity or its designee of the MIL-STD-2067 required reliability and maintainability analyses.

3.10.1 Mock-up, proofing and field maintainability review article evaluations. Escape system mock-up, engineering proofing article, field maintainability review article and production proofing article shall be subjected to a detailed design examination/critique conducted by the Government procuring activity, or its designee, in accordance with AR-105 to ensure (a) compliance with the applicable design specification, (b) utilization of good design practices and avoidance of bad practices, and (c) design benefit from reported fleet experience and problems.

3.10.2 Design evaluation program flow sequence. Figure 15 Aircrew Automated Escape System Design Evalutation Program Event Flow Sequence (page 81) is provided to illustrate the time and general relationships between the program major elements and phases.

3.11 Government witnesses. The contractor shall notify the Government procuring activity and/or its designee sufficiently in advance of each test to permit a Government witness to attend. "Test Witness Guidelines and Checklist" are provided in Appendix A for information concerning the duties and responsibilities of Government witnesses.

3.12 General problem areas. Appendix B furnishes comments and guidance concerning a number of problem areas which have impacted past escape system test programs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sled inspections. Following each test run, the sled shall be inspected to ensure its structural integrity for subsequent test usage.

4.2 Criteria for judging the acceptability of the results of individual tests and test programs. The criteria specified in this section have been established to provide assistance in determining the acceptability of the results obtained from both the individual test of the aircrew escape system and from the total test program. As much as is possible, the deter-

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

mination of the acceptability of the individual tests and of the test programs shall be made on the basis of these criteria. On the basis of these criteria, the contractor shall propose, in the test reports required by 3.6, test classifications for review by the Government procuring activity.

4.2.1 Classification of individual tests. In accordance with the criteria provided herein, tests shall be classified as:

(1) No test.

(2) Failure.

(3) Acceptable.

This classification shall be entered with other test data in the matrix described herein.

4.2.1.1 No test. A "no test" shall be determined to have occurred if:

(1) The test vehicle speed was not within the acceptable speed and acceleration ranges (±5 percent of target speed or ±10 knots whichever is greater and ±1 g) for the test run from the time the escape sequence was initiated until the time that the seat/dummy is ejected from the test vehicle.

(2) The escape system fails to operate as a result of:

(a) A non-escape system (i.e., sled, test initiation system) malfunction, or

(b) An escape system malfunction caused by maintenance error. (Malfunctions of the escape system caused by maintenance errors may be sufficient grounds for requiring design changes to preclude similar in-service maintenance errors.)

(3) A failure in either, or both the data collection system or the photographic coverage, results in insufficient data or coverage to permit adequate and satisfactory analysis of the test.

(4) Unauthorized personnel disturbing/damaging test articles during post-test recovery and/or analysis phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

c. Failure. A test shall be determined to be a failure in the event any of the following, or similar conditions occur during the test:

(1) An escape system failure caused by other than a non-escape system malfunction or a maintenance error.

(2) Ejection fails to occur within acceptable time limits.

(3) The rocket(s) (when part of the ejection sequence) fail(s) to ignite in the proper time sequence.

(4) The booster and/or rocket fails to propel the escape system to the required minimum velocity.

(5) The dummy's restraints and personnel equipment connections to the aircraft fail to release prior to or during the dummy/seat separation phase of the escape.

(6) A seat, during exposure to windblast prior to or during the dummy/seat separation phase, fails in such a manner as would be highly hazardous to a human occupant or causes a system malfunction resulting in a failure to properly eject and recover the dummy.

(7) The escape system propulsion and/or system pyrotechnics cause burn damage to either the escape system or seat/man stabilization or the recovery parachute which would impair normal function of these subsystems.

(8) The drogue parachute becomes entangled with the seat.

(9) The capsule or dummy fails to clear the projected aircraft structure by an adequate safety margin. (Safe clearance shall be demonstrated by drawing to appropriate scale an 8 foot diameter circle about the dummy and centered at the center of the dummy. If the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

circle clears aircraft structure projections, the clearances shall be presumed safe. A similar system will be developed for capsule systems.)

(10) The parachute fails to deploy.

(11) The parachutes fail to inflate with an adequate clear drop (descent while hanging vertically under a fully blossomed parachute) following actuation of escape system at ground level, for a fully equipped 98 percentile dummy or cockpit capsule.

(12) Structurally, parachutes must withstand opening forces with only minor damage. Operationally, parachute system and components must perform sequentially as designed within tolerances set for operational envelope of escape system. The personnel parachute sustains damage such that:

(a) The upper or lower lateral band(s) is broken (i.e., torn/cut or burned through) or otherwise damaged during the escape sequence.

(b) The parachute, including suspension lines and risers, sustains sufficient damage so that the dummy's steady state descent rate at the time of touchdown exceeds 30 fps total velocity and or a vertical component of 24 fps.

(c) The development time (from pack opening to full open parachute) exceeds the anticipated (calculated or previous actual test results) time by 50 percent of the normal development time for the velocity of the test.

(13) Occurence of line-covers or full inversion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

(14) Inadequate clearance between the capsule or dummy, including attached equipments and/or attached escape system elements, and other elements of the escape system, e.g. other ejected seats, cockpit canopies, own seat after separation, etc. (Safe clearance shall be demonstrated by drawing about the capsule and its attached equipments and/or attached escape system elements, or about the dummy and its attached equipments and/or attached escape system elements an envelope extending from the capsule or dummy center of gravity or, if applicable, extending from the axis of the capsule, or dummy, and attached equipments and attached escape system elements to a diameter at least one hundred percent greater than the actual diameter at that point (e.g.: an 8 feet diameter circle centered at the center of gravity of a dummy). Penetration of the envelopes so defined shall be considered demonstration of inadequate clearance.)

(15) Operation of the escape system produces one or more conditions classifiable as Class I or Class II hazards (MIL-STD-882).

(16) Human physiological/biomechanical injury threshholds are exceeded.

4.2.1.3 Acceptable. A test of the system shall be judged to have been satisfactory if (1) cockpit capsule or dummy and ejection seat system have been ejected from the test vehicle along a trajectory insuring a safe margin for clearing the aircraft structure and sufficient clear drop following blossoming of the parachute, (2) there have been no escape system malfunctions as described heretofore, and (3) there have been no test system/instrumentation or photographic coverage failures as previously described.

4.2.2 Satisfactory test program results. Satisfactory completion of a minimum of 80 percent of the tests, of which at least 8 acceptable Table II or Table III tests shall be consecutive, of the qualification program shall be considered acceptable, providing no failures occur and the remaining 20 percent of the tests are "no test" tests in which the escape system operated acceptably. However, a "no test" involving two or more

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

of the above failures may not be accepted. (See (f)(1)c.) No more than one test in a test program reduced to eight or fewer tests may be a "no test" as defined by categories (1), (2) and (3) below:

(1) The test vehicle speed was within ±15 percent of the target test speed, or

(2) The data collection was inadequate, or

(3) The photographic coverage was inadequate.

4.2.3 Conditions requiring re-running individual tests. Under the following conditions tests of the escape system shall be re-run:

(1) No tests.

(a) Those classified as "no test" as a result of an escape system malfunction induced by maintenance error.

(b) Those classified as "no test" as a result of failure of the escape system to operate, as a consequence of a non-escape system malfunction.

(c) Those classified as "no test" in which either two or more causes for the "no test" classification are involved, or in which the test vehicle speed during escape exceeded the target speed by ± 15 percent or 60 kts whichever is the lesser, unless the Government procuring activity considers the test data, photographic coverage, and test conditions satisfactory when considered with results obtained from the remainder of the test program.

(2) Failures. All tests the Government procuring activity classifies as "failures" shall be re-run. If the contractor deems it necessary to modify the escape system to insure success under the test conditions, the Government procuring activity shall be informed as to the nature of the modifications made. Further service release testing shall be deferred

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

until the modifications have been (1) defined, developed and demonstrated to a degree sufficient to satisfy the Government procuring activity that the proposed modifications can correct the problem and (2) the full modification has been incorporated in the test articles. If the Government procuring activity considers the modified escape system to be significantly changed, additional escape system tests may be required. In the event the contractor considers no modifications are required, further testing shall be deferred until the Government procuring activity has been satisfied concerning the soundness of the contractor's position and supporting rational and evidence.

(3) Non-Consecutiveness of successful tests. In the event upon completion of the approved service release test program (3.8.4.3.2.1) the successive successful tests fail to meet the following:

No. of Tests in Program	Minimum No. of Successive Successful Tests
6	6
7	6
8	7
9	8
10 or more	8

Such additional tests as are required to achieve the minimum number of successive successful tests shall be conducted.

(4) Non-demonstration of system reliability. In the event upon completion of the approved service release test program (3.8.4.3.2.1) the results do not demonstrate achievement of the required level of reliability, such tests as are required to demonstrate achievement of the required level of reliability shall be conducted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

4.2.4 Criteria for outright system rejection. When system as a whole fails to perform as planned and designed or when some component repeatedly fails without any practical method available to overcome the discrepancy, the Government procuring agency may reject the escape system.

5. PACKAGING

5.1 Not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended to prescribe uniform procedures to be followed in subjecting the components of aircraft emergency escape systems to tests to determine the adequacy of the system in providing escape of crewmen from aircraft in flight.

6.2 Function of data requirements. The data required by this specification and its subordinate specifications and standards has been specifically requested in order to:

(1) Cause the generation within the contractor's organization of expert knowledge concerning the contracted system.

(2) Identify at the earliest practicable dates within the program, all potential/possible problem areas.

(3) Cause contractor corrective action(s) where necessary to ensure fullest compliance with contract/specification requirements.

(4) Enable the Government procuring activity to gauge the validity and effectiveness of each required contractor effort to ensure contract/ specification compliance.

(5) Re-assure the Government procuring activity that either all is well or that the problems are identified and the risks knowingly accepted or that action has been initiated to resolve the problems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

Two prime concerns govern data delivery dates:

(1) Ensuring data timeliness sufficient to permit the most economic corrective action accomplishment where required to ensure fullest compliance with contract/specification requirements and, also to ensure timely availability of data necessary for the safe, effective, economic conduct of each phase of the program.

(2) Avoidance of excessive front-end labor-loading under the contract which, although it will produce the required plans, analyses and reports, never adequately enhances the longterm corporate knowledge and thus the program in the manner intended since the analysts, preparers and writers quickly leave the program and/ or company.

6.2.1 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph/Section	Data Requirement	Applicable DID
3.1.1	Test article non-conforming parts report	
3.1.2	Test article component re-use list	
3.1.3	Test article CFE/GFE requirements list	
3.3.4	Test frequency allocation application	DI-R-2067/ AE-001A
3.4	Test instrumentation data	DI-T-1908
3.4, 3.5, 3.6, 3.9	Test Reports	
3.4.2, 3.4.3	Photographic coverage plan	
3.4.2	Photographic coverage reports	
3.5	Marginality of success plan	
3.5	Test scheduling report	DI-T-1909
3.5.1	Test plan change proposal	DI-T-26391
3.5.2	Government facility test support document	
3.5, 3.8.1	System/subsystem/component test plans and specifications	
3.5, 3.8.3.4	Electromagnetic compatibility test plan	UDI-T-21331

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

Paragraph/Section	Data Requirement	Applicable DID
3.5, 3.8.3.4	Electromagnetic interference test plan	DI-R-2065/R020
3.6	Marginality of success reports	
3.6.2	Preliminary report of test results	DI-T-1905
3.6, 3.8.3.4	Electromagnetic compatibility test report	UDI-T-21332
3.8.1, 3.8.4.3.4.1.1	Request for approval for reduced testing of AAES and/or AAES elements	
3.8.2	IDCIA chits (blank)	
3.8.3	Environmental conditioning results data	
3.8.4.3.3.1	Ballasting data	
3.8.4.3.4.1.2	Test program	
3.9	Performance analyses and charts	

(Copies of data item descriptions required by the Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Preparing activity:
Navy - AS
(Project No. 1680 N446)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE I. Seat/man restraint physiological acceptability tower test program.
(reference paragraph 3.8.4.3.4.1)

Part A

Subject	Percentile Size	1/ "G" Level	2/ No. tests
Dummy	3	8	1
Dummy	98	8	1
Dummy	3	12	1
Dummy	98	12	1
Dummy	3	16	1
Dummy	98	16	1
Dummy	3	20	1
Dummy	98	20	1

3/
Part B

4/ Subject	Percentile 5/ Size	1/ "G" Level	2/ No. tests
Human	3	6	1
Human	50	6	1
Human	98	6	1
Human	3	8	1
Human	50	8	1
Human	98	8	1

(Continued next page)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE I. Seat/man restraint physiological acceptability tower test program.

Part B (Continued)

Subject 4/	Percentile 5/ Size	"G" Level 1/	No. tests 2/
Human	3	10	1
Human	50	10	1
Human	98	10	1
Human	3	12	1
Human	50	12	1
Human	98	12	1
Human	3	14	1
Human	50	14	1
Human	98	14	1
Human	3	16	1
Human	50	16	1
Human	98	16	1
Human	3	18	1
Human	50	18	1
Human	98	18	1
Human	3	20	1
Human	50	20	1
Human	98	20	1

(Continued next page)

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

TABLE I. Seat/man restraint physiological acceptability tower test program.
(continued)

1/ Peak "g" level to be that specified test "g" level which is immediately above the highest "g" level achieved during the high temperature tests of escape system propulsion subsystem using 3rd percentile aircrewman.

2/ In the event redesign of restraint subsystem and/or body support features is required at any stage, testing shall be stopped until redesign has been completed and incorporated. Redesign during Part A will require rerunning the tests starting at the "g" level preceding the level at which the problem occurred. If the redesign occurs in Part B, before Part B can be resumed, Part A must be rerun starting at the "g" level preceding the level at which the problem occurred.

3/ Prior to commencing Part B, all dynamic crash tests (3.8.4.1) and all Part A, Table I tests shall have been completed satisfactorily.

4/ All human subjects must undergo a thorough pre-test medical examination. Following each test the subjects should be questioned concerning, and examined for evidence of, contusions, pain, strain, and/or other injury.

5/ Closest approximation of weight, stature, sitting height, and buttock-knee length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE II. Track test program (single crew station system). 1/
(reference paragraphs 3.6.1, 3.6.3, 3.8.4.3.2.2, 3.8.4.3.3.1, 3.8.4.3.4, 3.8.4.3.4.1, 3.8.4.3.4.1.1, 4.2e)

Speed (knots)	Percentile Anthropomorphic Dummy	No. of Tests
2/ 0	3	2
2/ 0	98	2
3/ 50	3	2
3/ 50	98	2
100	3	1
100	98	1
265	3	1
265	98	1
360	3	1
360	98	1
435	3	1
435	98	1
500	3	1
500	98	1
4/ 600	3	2
4/ 600	98	2

1/ For escape systems incorporating sensor and control subsystems to select the mode of escape system operation, four (4) additional tests will be required: Two (2) at 25 kts below the crossover speed and two (2) at 25 kts above the crossover speed. For these four (4) tests the sensor and control subsystem shall be locked into the desired mode. In no other tests shall the sensor and control subsystem be locked into the desired mode.

2/ At most adverse ejected seat-man system center of gravity location.

3/ At most adverse ejected seat-man system center of gravity location with one test at 20 degrees nose down and one at 30 degrees bank.

4/ Or maximum aircraft speed, if less than 600 knots.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

TABLE III. Track test program.

(Multi-crew station systems and cockpit capsule systems). 1/
(reference paragraphs 3.6.1, 3.6.3, 3.8.4.3.2.2, 3.8.4.3.3.1, 3.8.4.3.4, 3.8.4.3.4.1, 3.8.4.3.4.1.1, 4.2e)

Speed (knots)	No. of tests
2/ 0	2
3/ 50	2
100	1
265	1
360	1
435	1
500	2
4/ 600	2

1/ Each test to use a mix of 3rd and 98th percentile dummies. The mix and location of the dummies shall be selected on the basis of system trajectory analyses predictions for achieving minimum clearances between systems and shall be subject to the approval of the Government procuring agency. For escape systems incorporating sensor and control subsystems to select the mode of escape system operation, two (2) additional tests will be required: One (1) at 25 kts below the crossover speed and one (1) at 25 kts above the crossover speed. For these two tests the sensor and control subsystem shall be locked into the desired mode. In no other tests shall the sensor and control subsystem be locked into the desired mode.

2/ At most adverse ejected seat-man system center of gravity location.

3/ At most adverse ejected seat-man system center of gravity location with one test at 20 degrees nose down and one at 30 degrees bank.

4/ Or maximum aircraft speed, if less than 600 knots. For capsular-type escape system, the contractor shall propose additional tests to demonstrate capability of safe escape throughout the speed regime from 600 KEAS to VL.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Flight test program.
(single seat systems) 1/
(reference paragraphs 3.6.1, 3.8.4.3.4,
3.8.4.3.4.1, 3.8.4.3.4.1.1)

Speed (knots)	Attitude	Sink Rate	No. of Tests
2/	Level	2,400 fpm	2
2/	Inverted	0	2
2/	20° Dive	0	2
3/	Level	0	4

1/ These tests shall be performed using 98 percentile anthropomorphic dummies (except that 3 percentile dummies shall be used for the inverted test) and the most adverse ejected seat-man center of gravity location. In the event escape system configuration varies between crew stations, the test shall be repeated for each configuration.

2/ At low controllable flight speed.

3/ Tests are high altitude ejection tests to evaluate integrity of parachute subsystem following high dynamic pressure condition escape within the upper limits of the barometric control setting. Speed and altitude of test to be determined by contractor through analyses of man-seat drag, post-ejection man-seat speed decay until parachute opening, man-seat trajectory following ejection until parachute opening so as to ensure maximum opening shock/stresses in parachute. Following a personnel parachute failure, new ejection conditions shall be determined by the contractor in an attempt to bracket the safe parachute opening zone. All contractor proposed conditions shall be subject to Government procuring activity review and approval.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE V. Crash condition escape program. 1/
(reference paragraphs 3.6.1, 3.8.4.3.2.4, 3.8.4.3.4, 3.8.4.3.4.1

Speed 2/ (knots)	3/ Deceleration	No. of tests
80	+5g	2
80	+5g/2g/2g	2
80	-5g/2g/2g	2

1/ These tests shall be conducted using 3rd and 98th percentile anthropomorphic dummies ejected in seats positioned to simulate the most adverse escape system/man center of gravity.

2/ Speed at escape initiation.

3/ Dynamic crash acceleration loads on escape system along the X/Y/Z axes of aircraft.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Underwater escape tests.

3/

PART A

Canopy implosion depth test

To be conducted using a sealed cockpit section with production, or production equivalent, canopy(ies) properly mounted. The cockpit shall be lowered to a depth of fifteen feet (measured from the top of the canopy) and evacuated to simulate the pressure differential achieved by increasing depth until the canopy implodes. The pressure differential across the canopy and the simulated equivalent depth at which implosion occurs shall be recorded. If flooding doors are employed in system design, the flooding door/passages should be restricted in a manner simulating the actual installation and the test repeated in a manner simulating a 10 fps sinkrate.

3/

Part B

Underwater manual egress tests 4/

5/ Seat position	6/ Percentile size	Cockpit attitude	2/ No. of tests
Full down	3	Upright, level	1
Full down	98	Inverted	1
Full up	3	Nose down	1
Full up	98	Tail down	1

1/ Size dummy(ies) shall be selected to produce worst limitation upon escape capability under particular test conditions.

2/ Number of tests per aircrew station.

3/ CAUTION: Before human subjects may be used (Part C), canopy implosion test (Part B) must be completed to determine canopy configuration/test depth which may be utilized with SAFETY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

4/ To be conducted using trained SCUBA divers selected and equipped in accordance with 3.8.4.3.2.5.

5/ If tilt adjustment and/or fore-and-aft seat adjustment are incorporated in the seat, the tests shall be conducted at the most adverse combinations of adjustments.

6/ Closest approximation of weight and stature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Test report format.
(reference paragraphs 3.6.3, 3.6.5)

1. TITLE PAGE

   a. Company Name and Address

   b. Report Number

   c. Date of Release

   d. Author's Name, Title and Signature

   e. Reviewers' Names, Titles and Signatures

   f. Report Title

   g. Contract Title

   h. CDRL (DD Form 1664) Number

   i. DID (DD Form 1664) Number

2. ABSTRACT

   a. Synopsis of Testing Conducted (i.e., the number of tests conducted in the series covered in the report and the location of the test(s) covered in relation to the test series and test program).

   b. Synopsis of Findings (i.e., Brief statement describing the successes, anomalies and problems encountered during the reported tests).

3. TABLE OF CONTENTS

4. LIST OF TABLES

5. LIST OF ILLUSTRATIONS

6. TEST DESCRIPTION

(Continued next page)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE VII (Continued)

a. Test Purpose

b. Test Article (i.e., provide a complete detailed description of the test article configuration, including a description of each change from previous test article configuration with supporting rationale for each change).

c. Test Fixture

d. Data to be Acquired/Measured During Test

e. Instrumentation (i.e., Provide nomenclature, model number, part number, serial number, and calibration record for each piece of test instrumentation, including data transmission and recording systems, and provide a chart depicting camera locations citing for each the lens focal length, film speed, and type of film).

f. Test Procedures

g. Data Reduction and Analysis Techniques

h. Criteria for Acceptance/Rejection of Tests and Test Series

'. TEST RESULTS

a. Test Data Matrix

b. Test Data Graphical Presentations

(1) Single Test Smoothed Graphs

(2) Running/Trend Graphs Covering All Tests to Date for Specific Types of Data

(3) Parachute damage charts(Figures 2 and 3)

(4) Dummy damage chart (Figure 4 and 5)

(5) Personnel equipment/garment damage chart (Figure 6 and 7)

c. Failure/Marginality of Success Analysis

d. Analysis of Test Results

(Continued next page)

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:54Z
Check the source to verify that this is the currenversion before use.

TABLE VII (Continued)

e. Description of Test and/or Test-to-Test Anomalies

f. Description of Design Deficiencies

g. Test Evaluation in Accordance with Acceptance/Rejection Criteria with Supporting Rationale

8. CONCLUSIONS AND RECOMMENDATIONS

9. PROPOSED REMEDIAL ACTION

a. Description and Supporting Rationale

b. Proposed Incorporation Point Within Program

10. PROGRAM FOR NEXT INTERVAL

11. APPENDICES

a. List of Movie Films

b. List of References

c. Bibliography

d. List of Test Witnesses

e. Quality Assurance Records for Test Article

f. Disposition of Test Article

g. Instrumentation Records

h. Test Data Analysis Worksheets

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

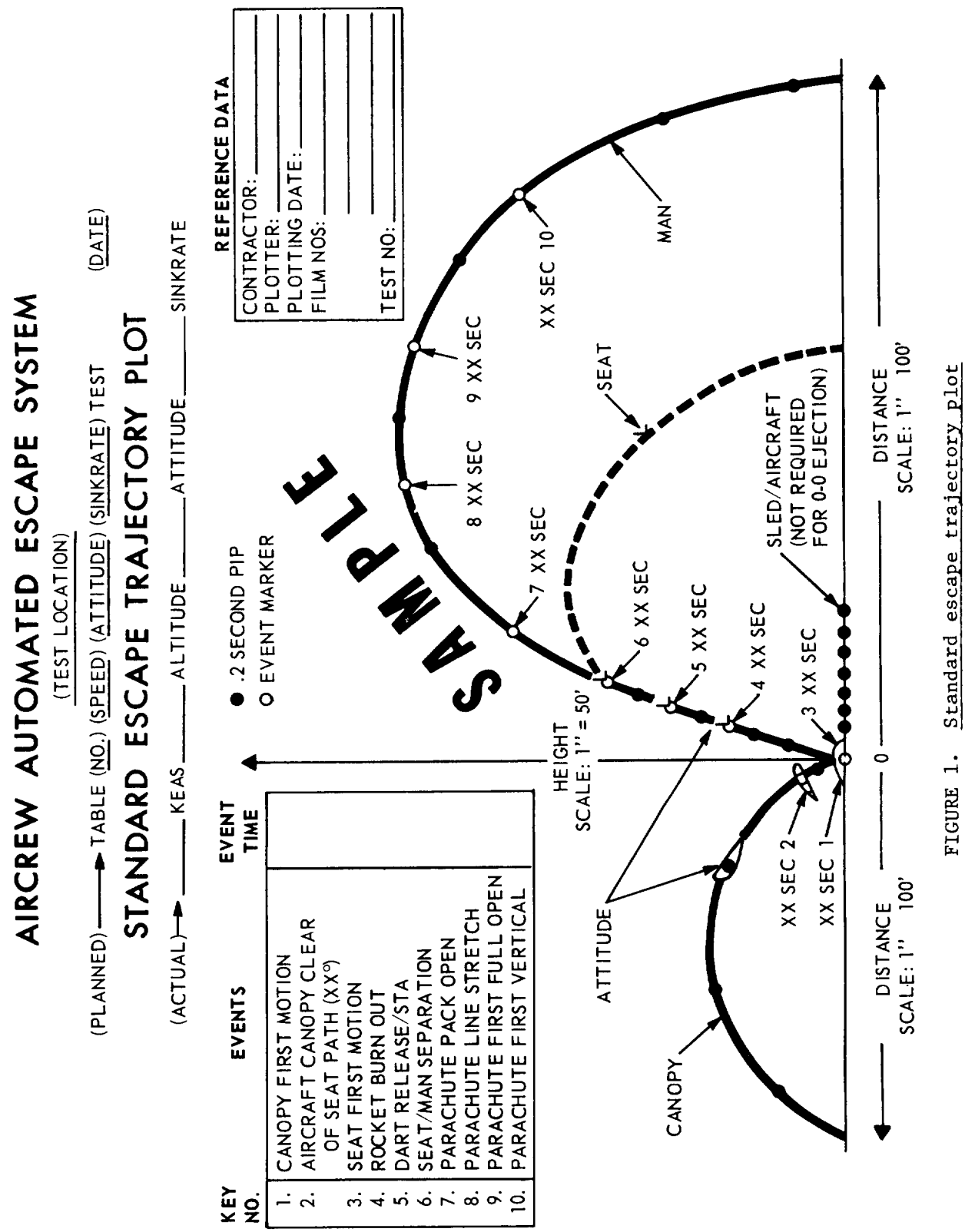


FIGURE 1. Standard escape trajectory plot

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

CANOPY DAMAGE CHART
— FOR 24—FOOT AND 28-FOOT FLAT CANOPIES—

ACTIVITY			DATE
DATE OF MANUFACTURE	CONTRACT NO.	SERIAL NO.	INSPECTOR (PRINT)

INSTRUCTIONS

1. USE FOLLOWING SYMBOLS TO INDICATE LOCATION AND TYPE OF DAMAGE ON CANOPY DIAGRAM.

6	TEAR, 6 — INCHES LONG
3 4	TEAR, "L" SHAPED WITH 3 — AND 4 — INCH LEGS
3 4	TEAR, "T" SHAPED WITH 4 — INCH BASE AND 3 — INCH LEG
8	RUPTURE, 8 — INCH DIA AT WIDEST POINT
	BURN MARKS
	STRAINED AREA
	PULLED DRAG SURFACE THREAD
	SINGLE BROKEN DRAG SURFACE THREAD
#	WEAVE SEPARATION
S/D	STAINS / DISCOLORATION
3	BROKEN SEAMS STITCHES ALONG 3 INCHES OF SEAM
	PULLED OR LOOPED SEAM THREAD
	OTHER DAMAGE: EXPLAIN

2. MARK DAMAGE USING SYMBOLS TO SHOW AREA AFFECTED.
3. CANOPY DIAGRAM CONTAINS 28 GORES FOR CANOPIES WHICH HAVE 24, DELETE EXTRA GORES IN DIAGRAM BY MARKING THEM "N/A" NOTE THAT 28— AND 24— FOOT DIA FLAT CANOPIES HAVE 28 AND 24 GORES RESPECTIVELY.
4. AS DAMAGE IS REPAIRED ON EACH GORE, PLACE SYMBOL "OK" ADJACENT TO SKIRT AT END OF EACH GORE.
5. 24—FOOT DIA CANOPIES MAY HAVE NAMEPLATE ON GORE 1 OR ON SKIRT HEM AT GORE 1.
6. 28—FOOT DIA CANOPIES HAVE NAMEPLATE ON GORE 28.

REMARKS: _____

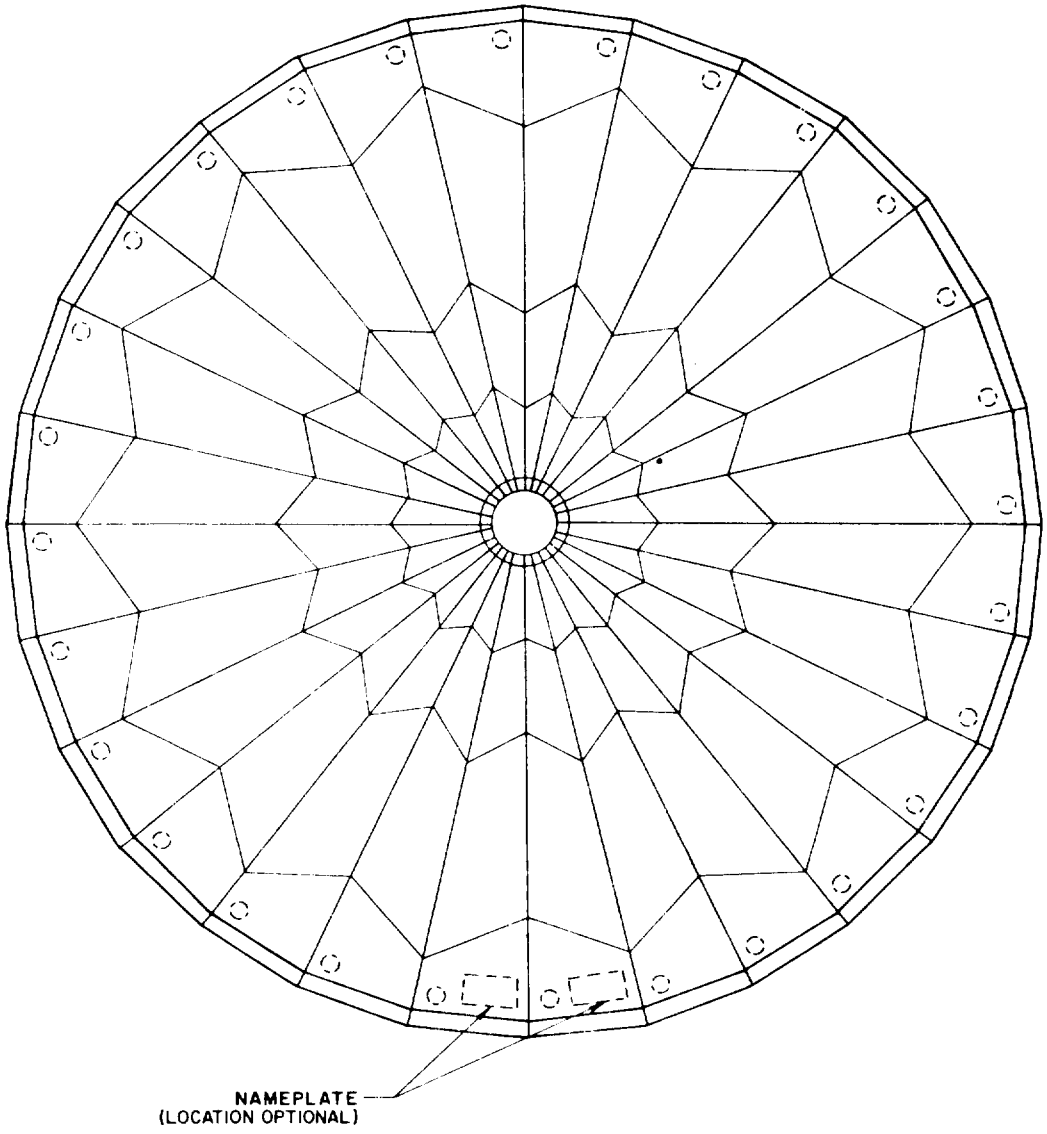
THIS IS A PERMANENT RECORD. DO NOT DISCARD.
UPON COMPLETION OF REPAIRS, THIS CHART IS TO BE AFFIXED TO THE PARACHUTE HISTORY CARD. (FRONT)

FIGURE 2. Sample canopy damage chart

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

CANOPY DAMAGE CHART (REAR)

CANOPY DIAGRAM (TOP VIEW)



THIS IS A PERMANENT RECORD. DO NOT DISCARD.
UPON COMPLETION OF REPAIRS, THIS CHART IS TO BE AFFIXED TO THE PARACHUTE HISTORY CARD.

FIGURE 3. Sample canopy damage chart

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

DUMMY DAMAGE CHART

ACTIVITY			DATE
DATE OF MANUFACTURE	CONTRACT NO.	SERIAL NO.	INSPECTOR (PRINT)

INSTRUCTIONS

1. USE FOLLOWING SYMBOLS TO INDICATE LOCATION AND TYPE OF DAMAGE ON DUMMY DIAGRAM.

6	TEAR, 6 — INCHES LONG
3 4	TEAR "L" SHAPED WITH 3 — AND 4 — INCH LEGS
3 4	TEAR "T" SHAPED WITH 4 — INCH BASE AND 3 — INCH LEG
8	RUPTURE, 8 — INCH DIA AT WIDEST POINT
	BURN MARKS

	ABRASIONS
S D	STAINS/DISCOLORATION
	AMPUTATED PART
	BREAK
	OTHER DAMAGE: EXPLAIN

2. MARK DAMAGE USING SYMBOLS TO SHOW AREA AFFECTED.

REMARKS: _____

(FRONT)

FIGURE 4. Sample dummy damage chart.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T12:54Z
Check the source to verify that this is the current version before use.

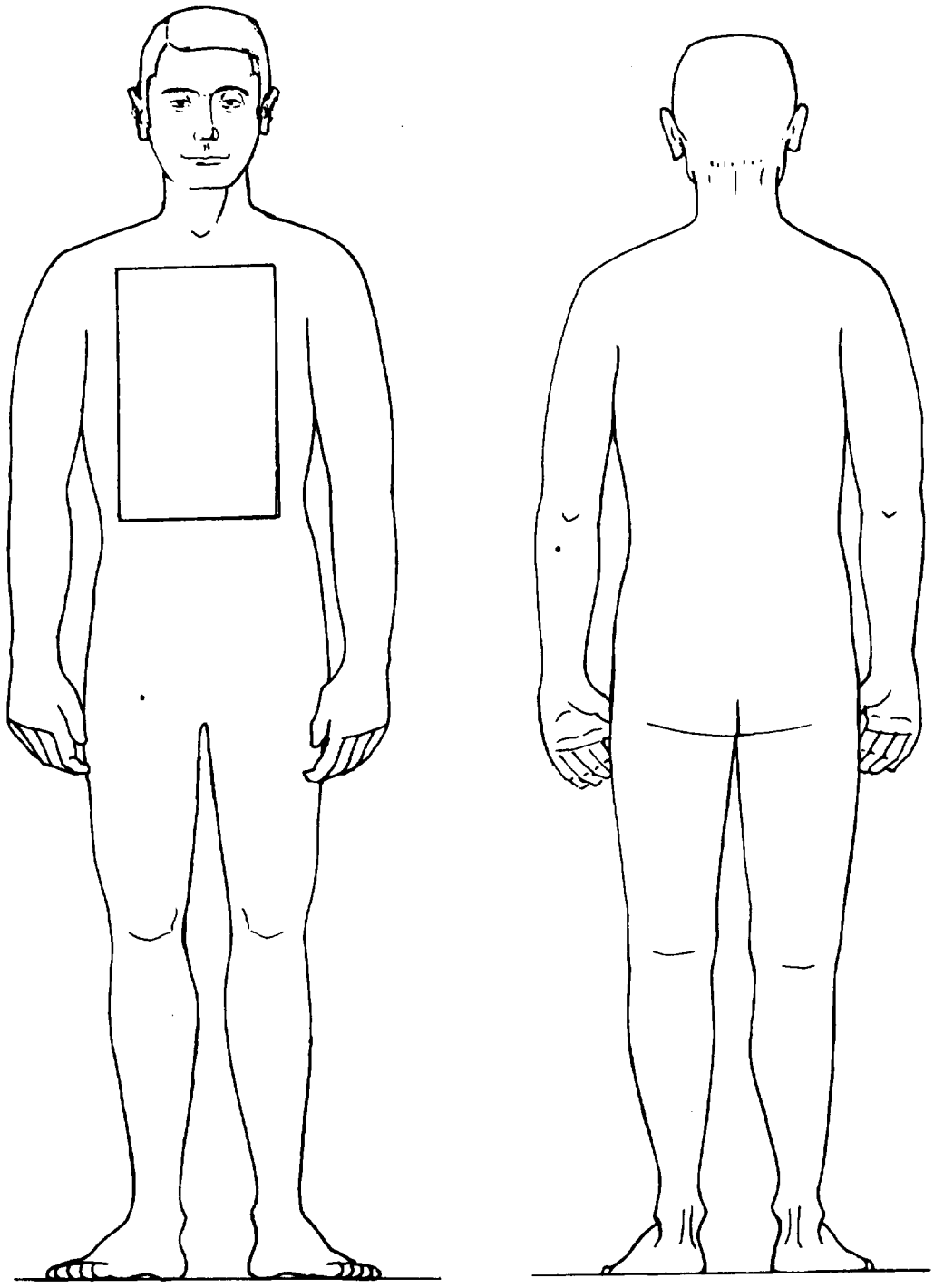


FIGURE 5. Dummy damage chart (sheet 1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

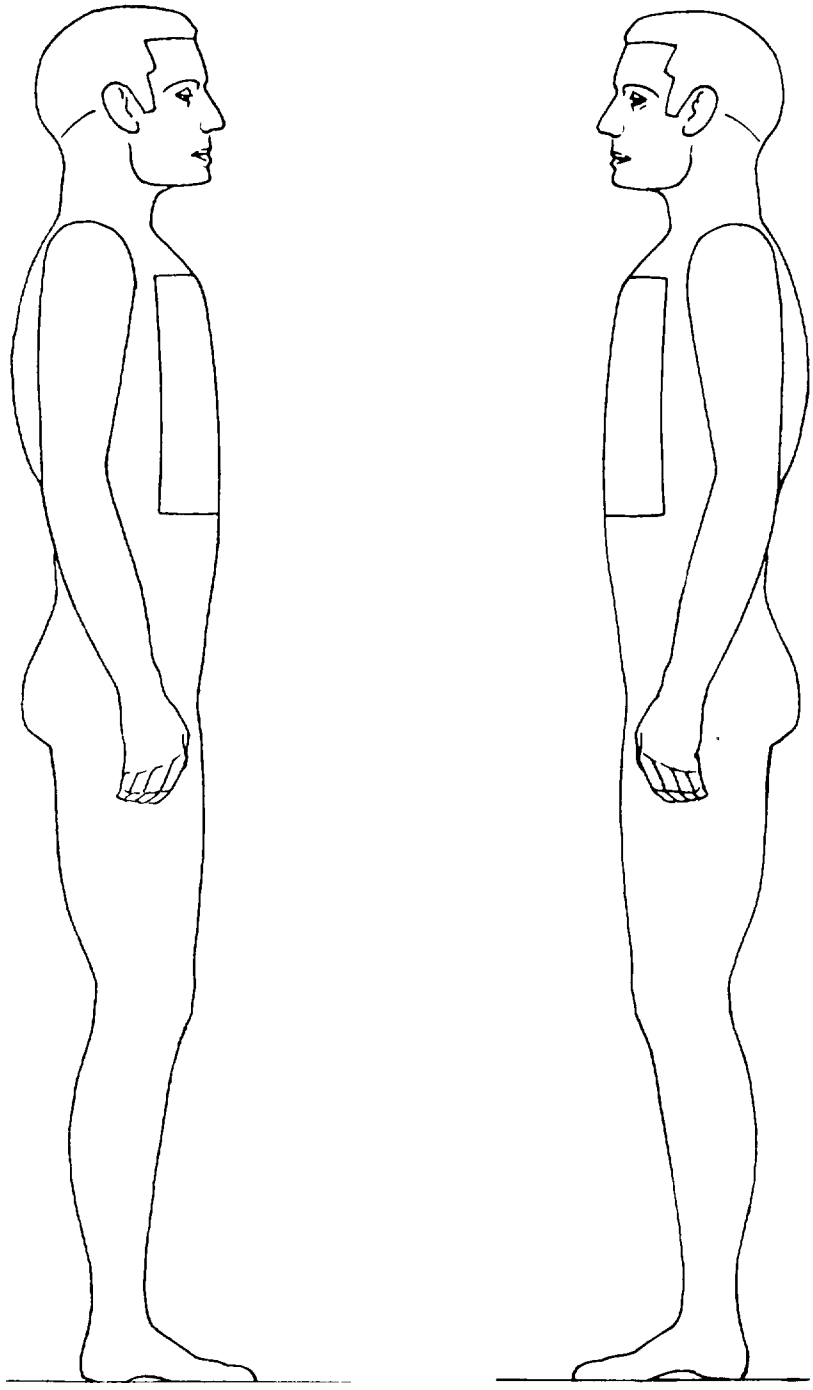


FIGURE 5. Dummy damage chart (sheet 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

EQUIPMENT DAMAGE CHART

ACTIVITY			DATE
DATE OF MANUFACTURE	CONTRACT NO.	SERIAL NO.	INSPECTOR (PRINT)

INSTRUCTIONS

1. USE FOLLOWING SYMBOLS TO INDICATE LOCATION AND TYPE OF DAMAGE ON EQUIPMENT DIAGRAM.

6	TEAR, 6 — INCHES LONG
3 4	TEAR "L" SHAPED WITH 3 — AND 4 — INCH LEGS
3 4	TEAR, "T" SHAPED WITH 4 — INCH BASE AND 3 — INCH LEG
8	RUPTURE, 8 — INCH DIA AT WIDEST POINT
	BURN MARKS
	STRAINED AREA
	PULLED DRAG SURFACE THREAD
	SINGLE BROKEN DRAG SURFACE THREAD

#	WEAVE SEPARATION
S/D	STAINS/DISCOLORATION
3	BROKEN SEAMS STITCHES ALONG 3 INCHES OF SEAM
	PULLED OR LOOPED SEAM THREAD
	SEPARATED PART
	BREAK
	OTHER DAMAGE: EXPLAIN

2. MARK DAMAGE USING SYMBOLS TO SHOW AREA AFFECTED.
3. COMPLETE ONE OF THESE SHEETS FOR EACH EQUIPMENT SKETCH COMPLETED.

REMARKS: _____

(FRONT)

FIGURE 6. Sample equipment damage chart

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

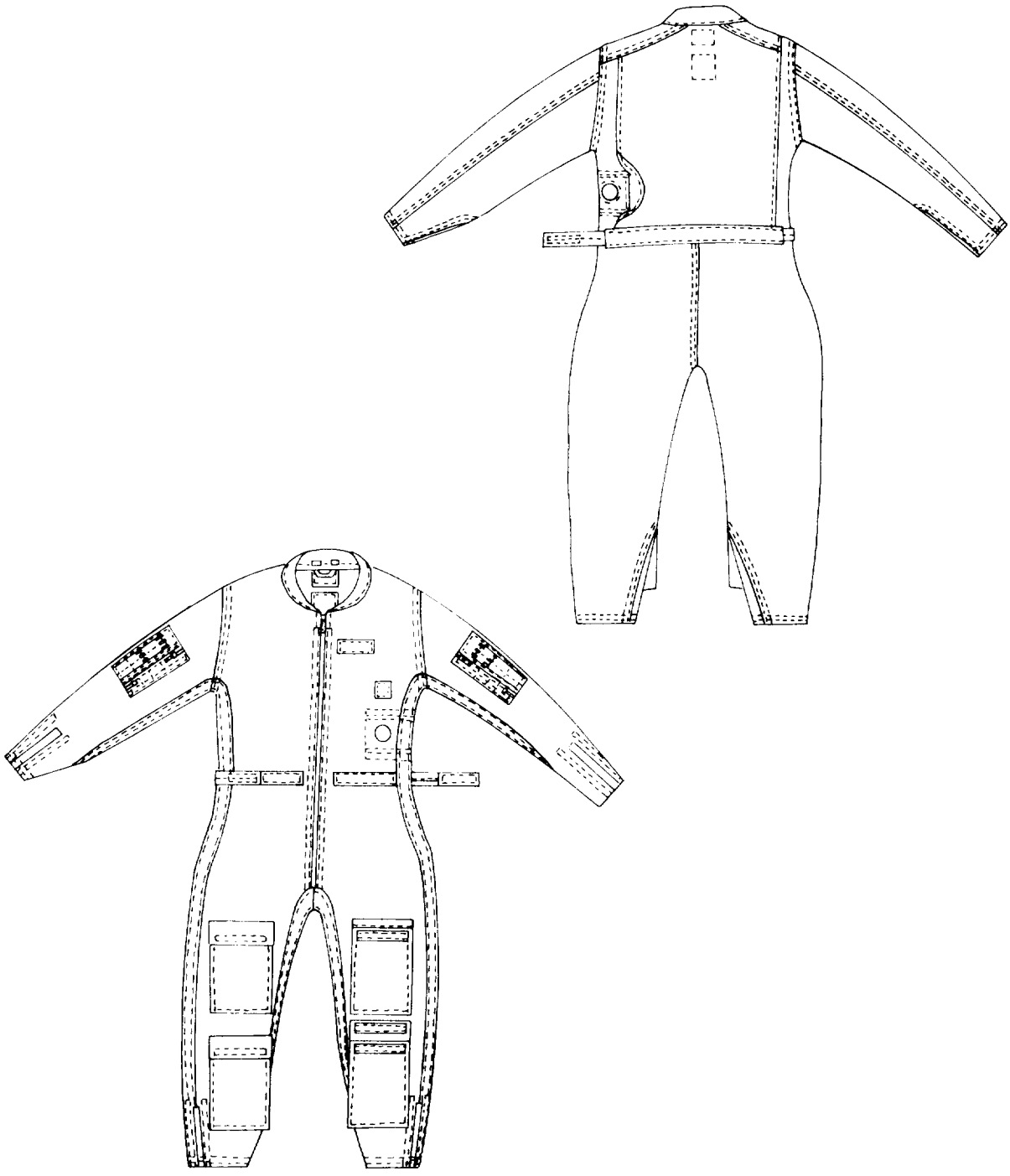


FIGURE 7. Equipment damage chart (sheet 1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

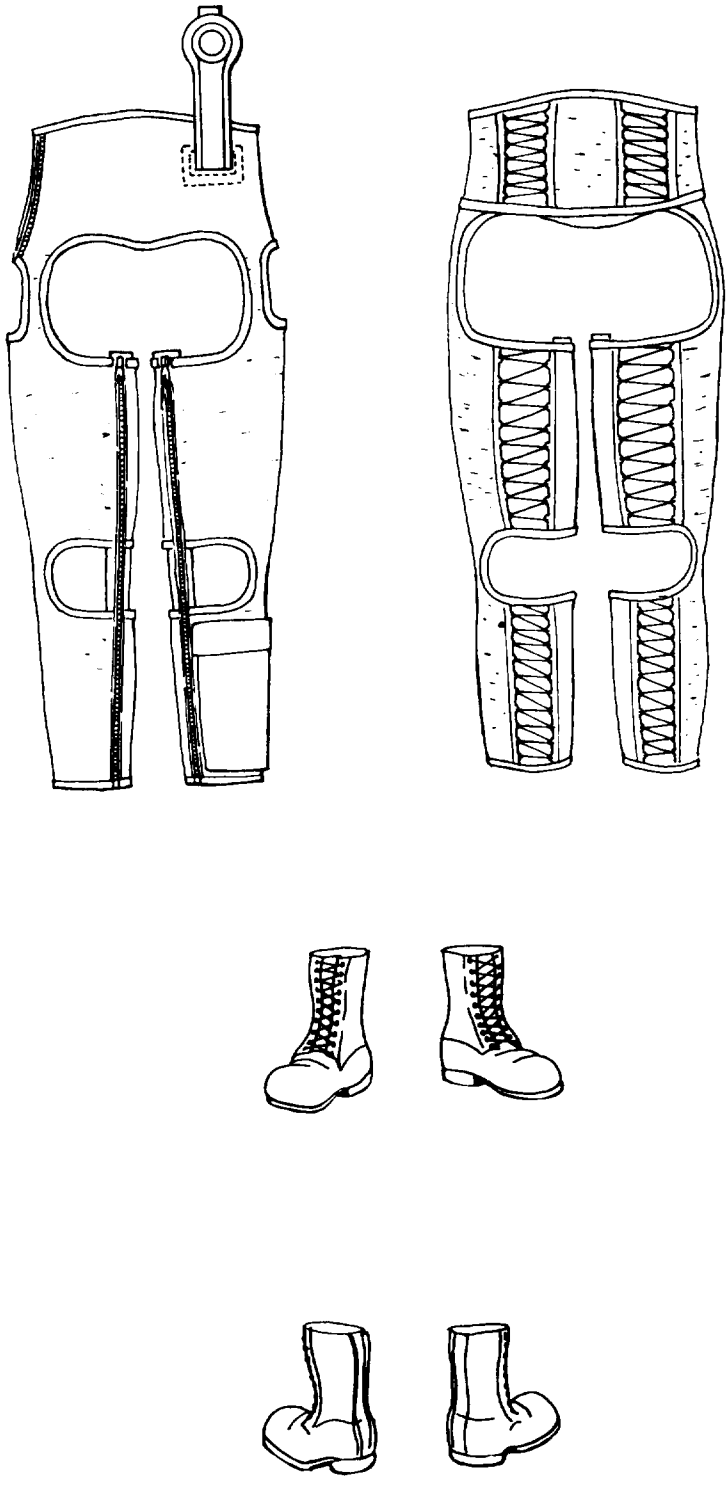


FIGURE 7. Equipment damage chart (sheet 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

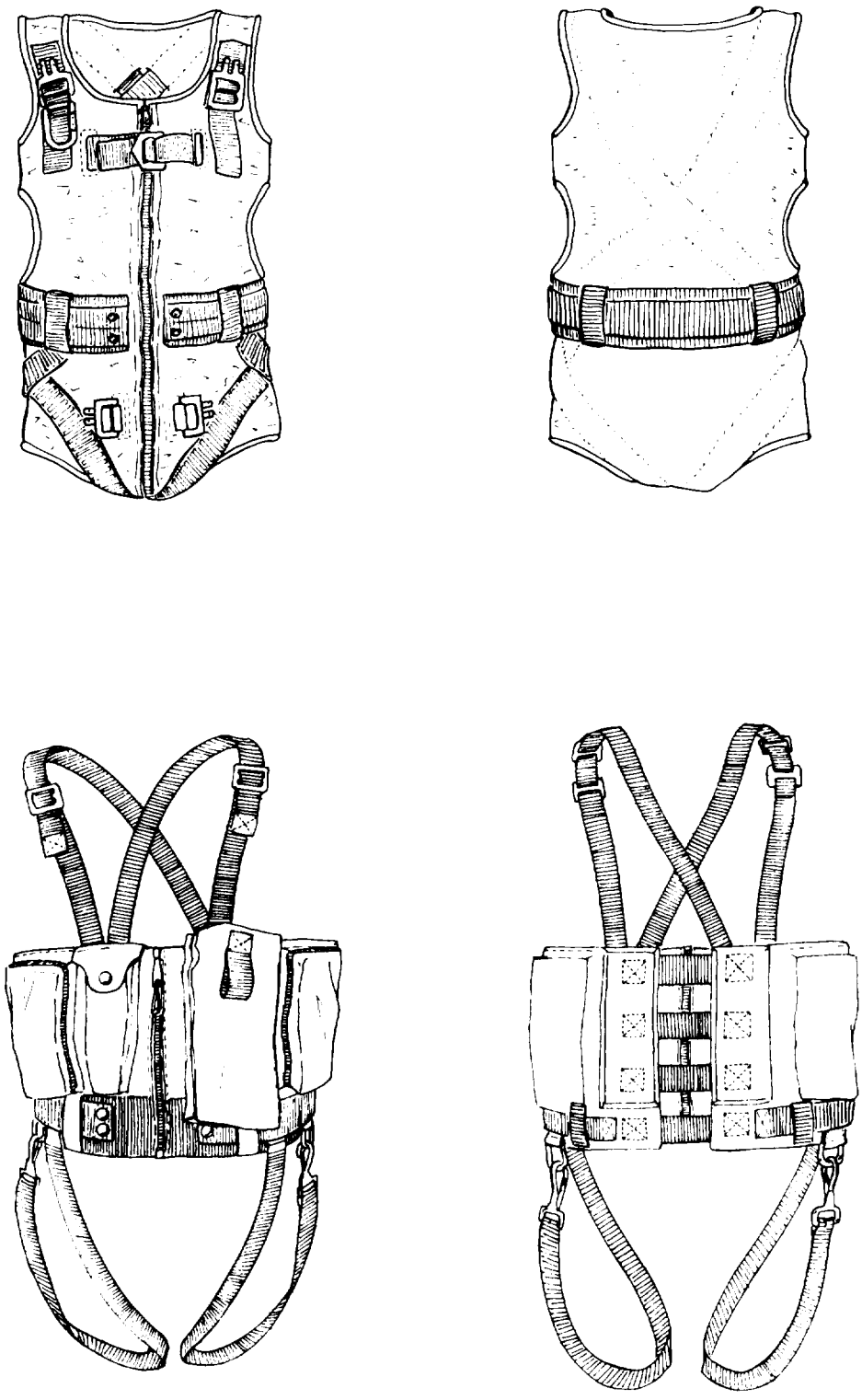


FIGURE 7. Equipment damage chart (sheet 3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

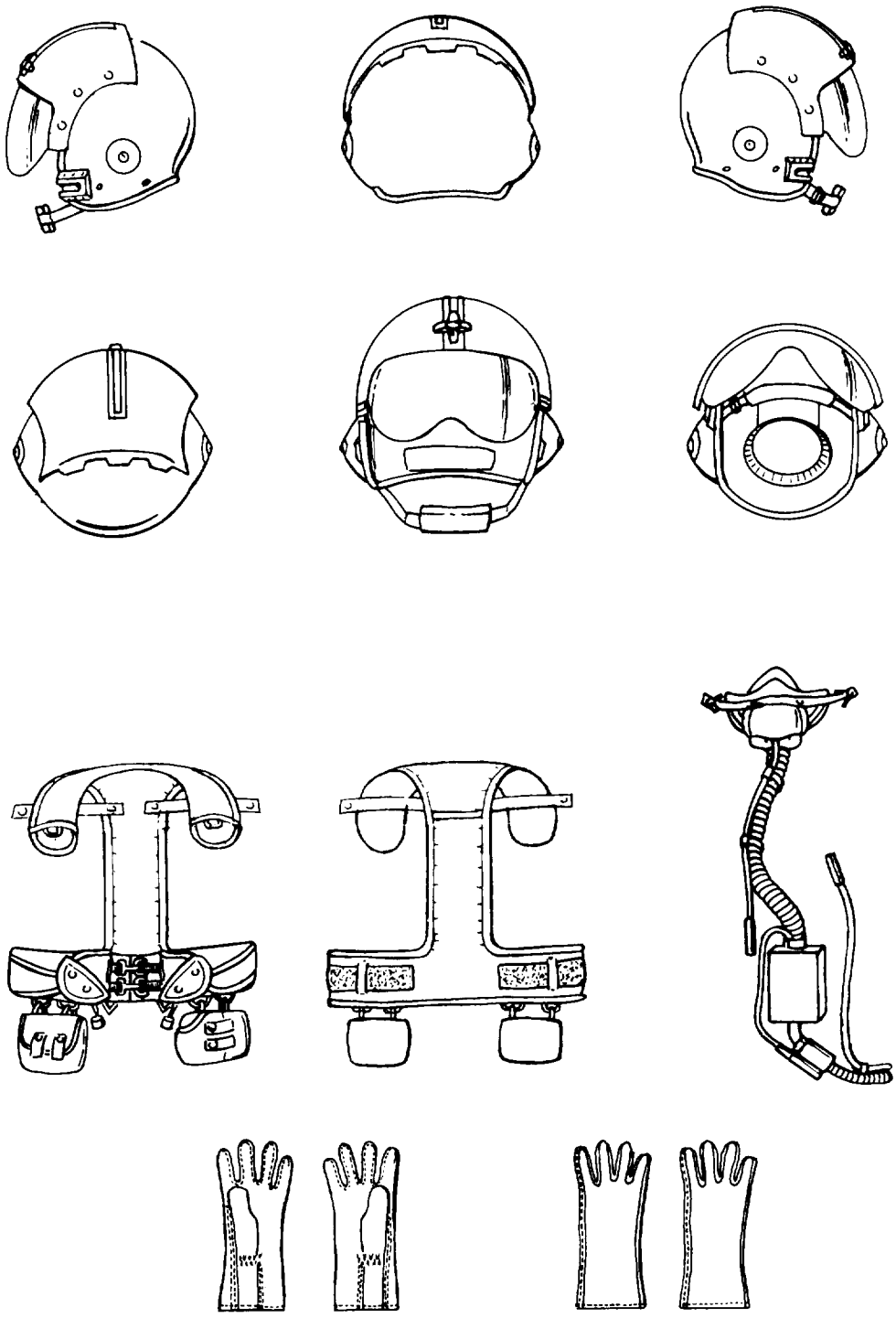


FIGURE 7. Equipment damage chart (sheet 4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

ESCAPE SYSTEM PERFORMANCE
AIRCRAFT PITCH ANGLE
VS
TERRAIN CLEARANCE ALTITUDE FOR SAFE ESCAPE

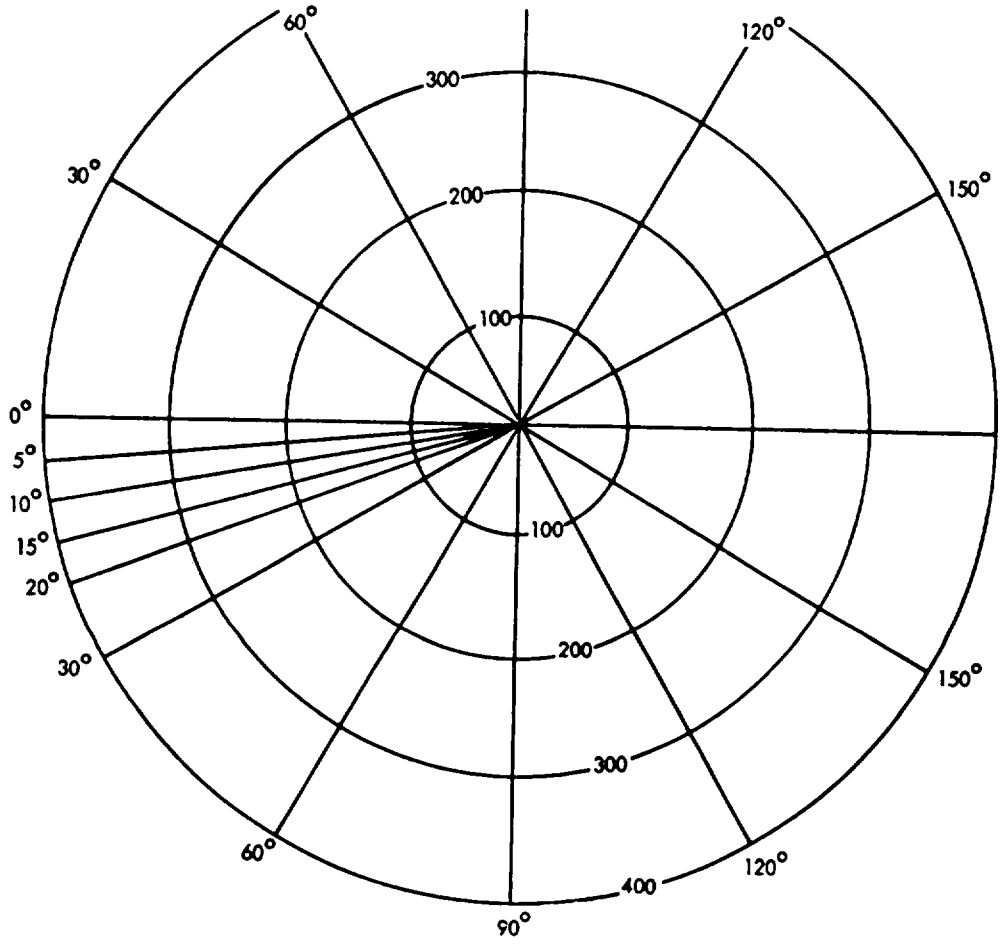


FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

ESCAPE SYSTEM PERFORMANCE
AIRCRAFT BANK ANGLE
VS
TERRAIN CLEARANCE ALTITUDE FOR SAFE ESCAPE

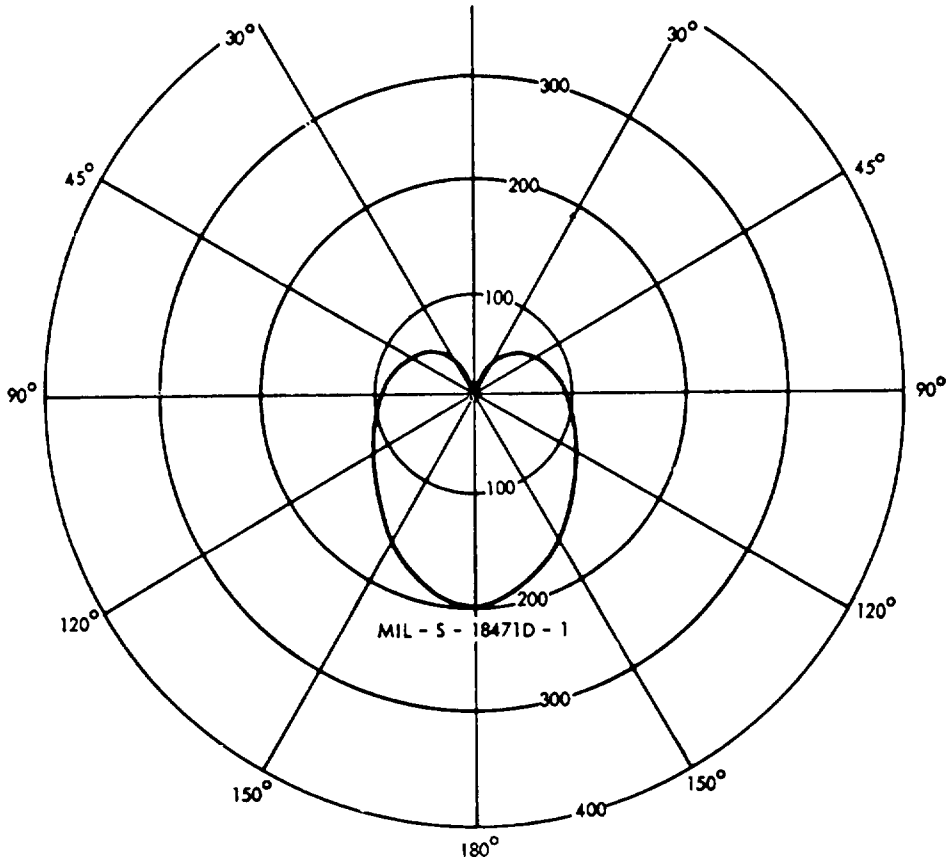


FIGURE 9

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

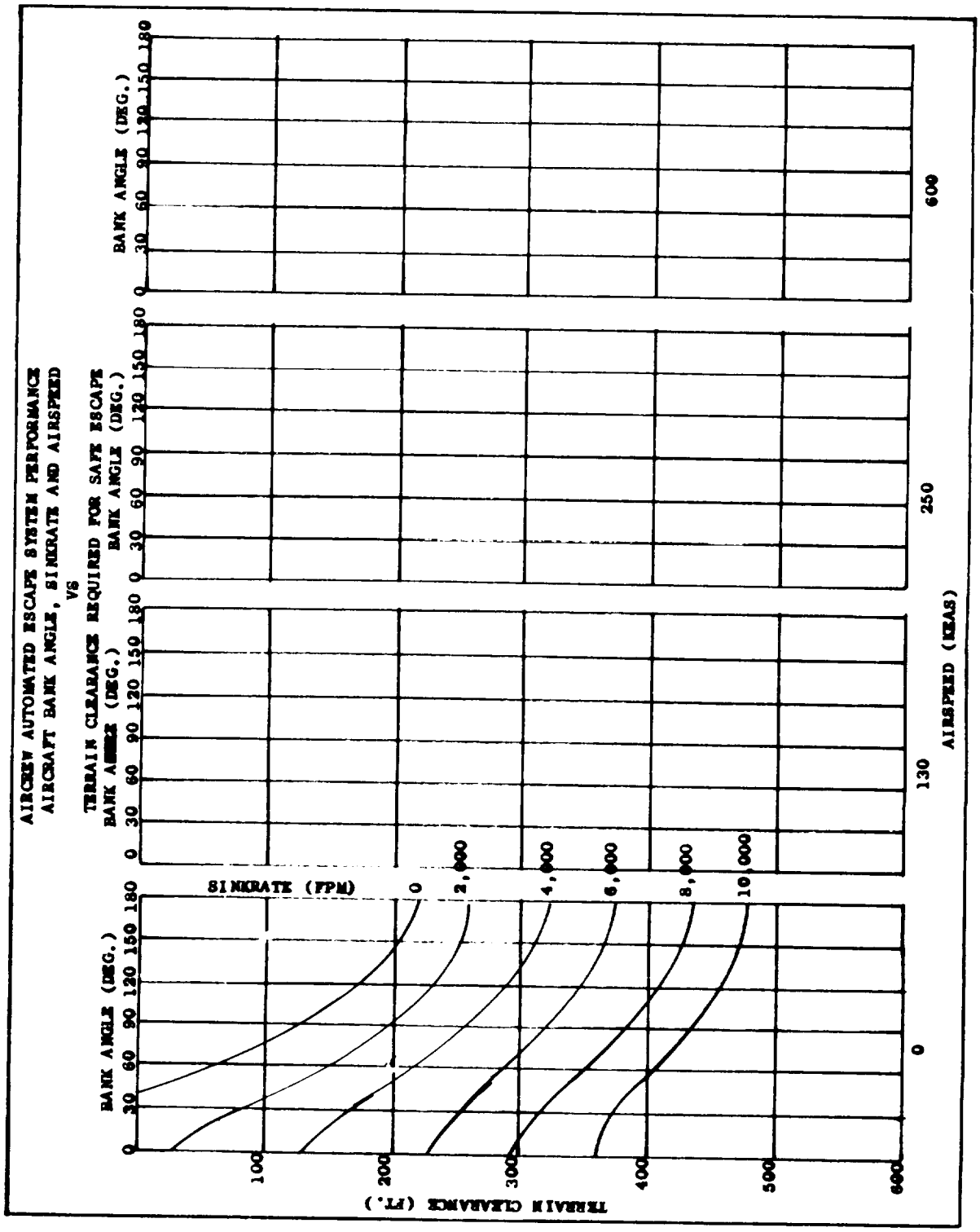


FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

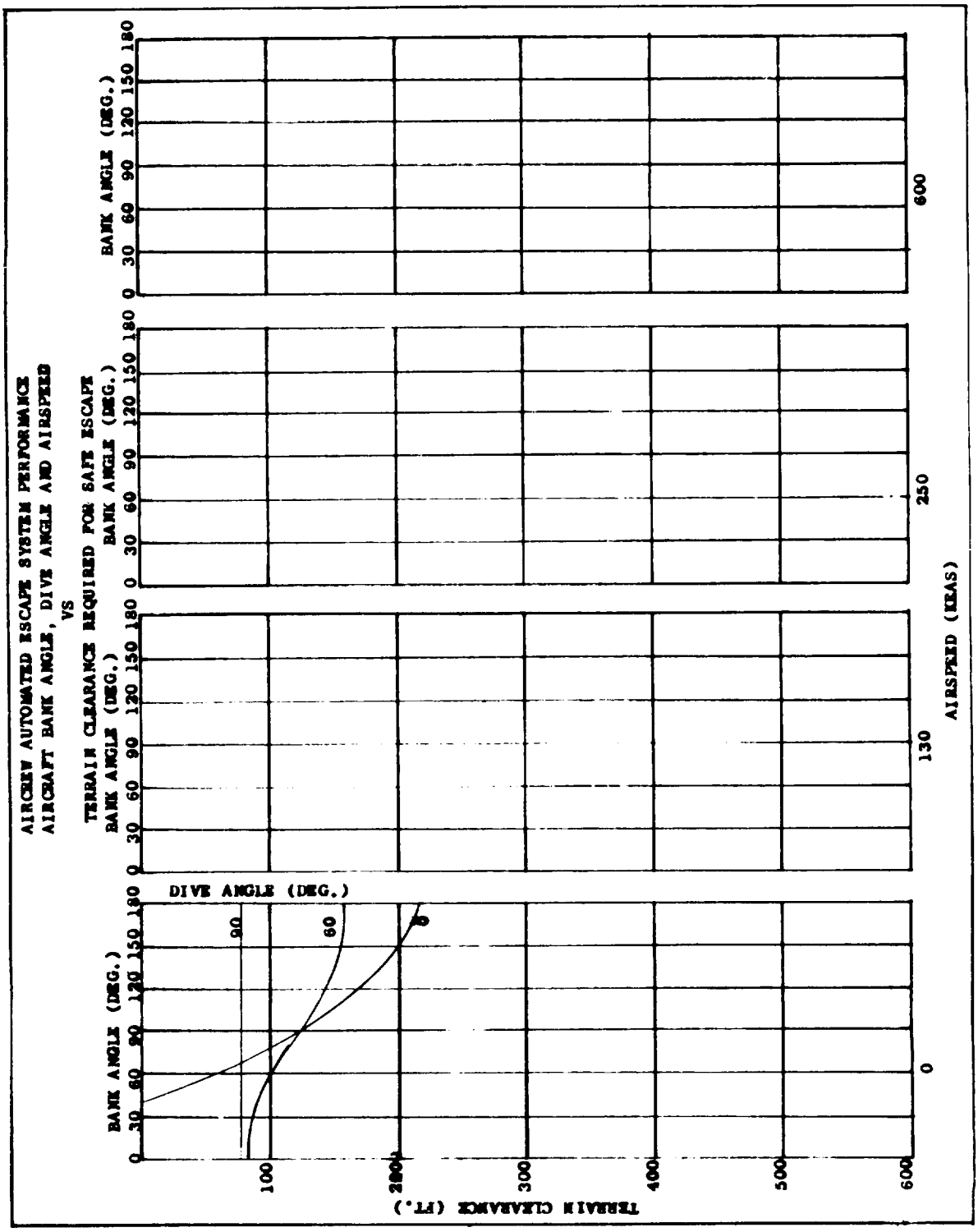


FIGURE 11

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

CHIT NO. _____

IN-PROCESS DESIGN CONFORMANCE INSPECTION FOR ESCAPE SYSTEM
FORM FORMAT

INSPECTION LOCATION: INSPECTION DATE: _____
ITEM INSPECTED: _____ REQUIREMENT REFERENCE (spec/para.): _____
SUBJECT:
DESCRIPTION OF DEFICIENCY: (Describe deficiency giving all pertinent part nos., pertinent specification references and effect upon system. If feasible, attach sketches or photographs illustrating deficiency and its effects/problems)
DEFICIENCY AFFECTS: SAFETY___RELIABILITY___MAINTAINABILITY_____ PERFORMANCE___ COCKPIT COMPATIBILITY___CREW ACCOMMODATION/ PERFORMANCE_____
RECOMMENDATION: _____ ORIGINATOR'S NAME & ORGANIZATION
CONTRACTOR COMMENTS:
NAVAIR: CONTRACTOR:

SHEET____OF____

FIGURE 12. In-process design conformance inspection for escape system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

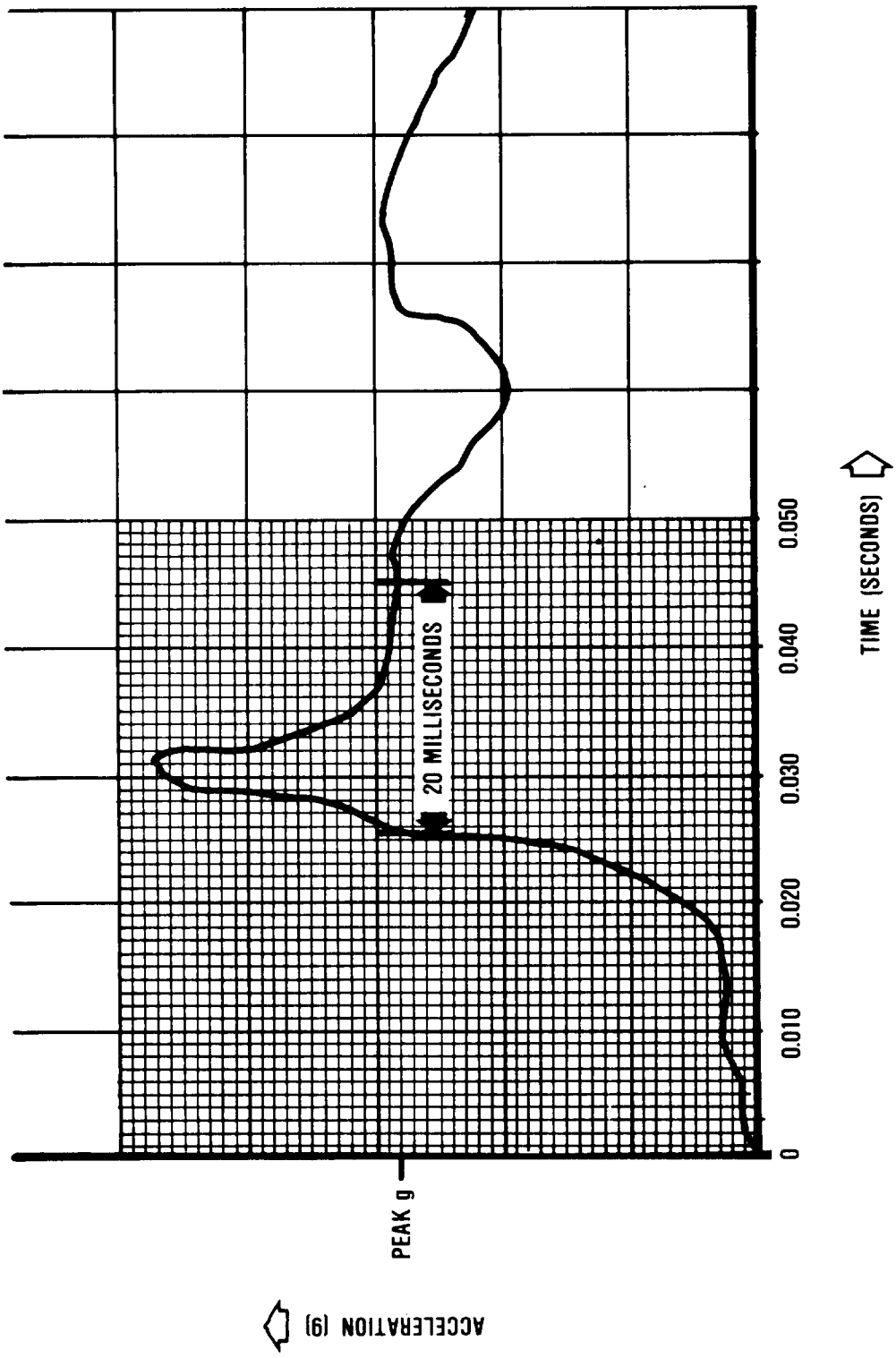


FIGURE 13. Sample acceleration trace depicting required technique for measuring peak acceleration

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

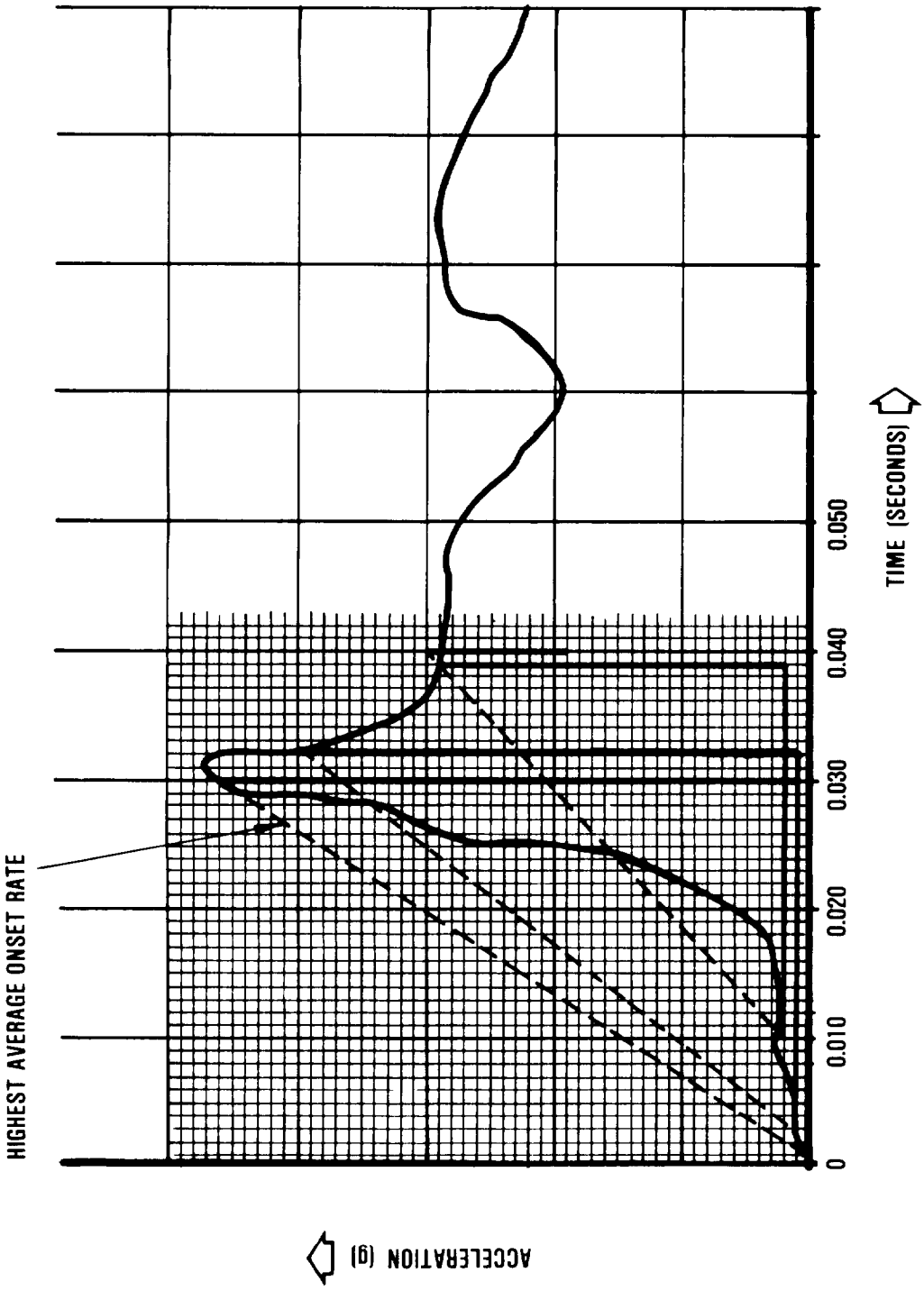


FIGURE 14. Sample acceleration trace depicting required technique for measuring highest rate of G onset

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

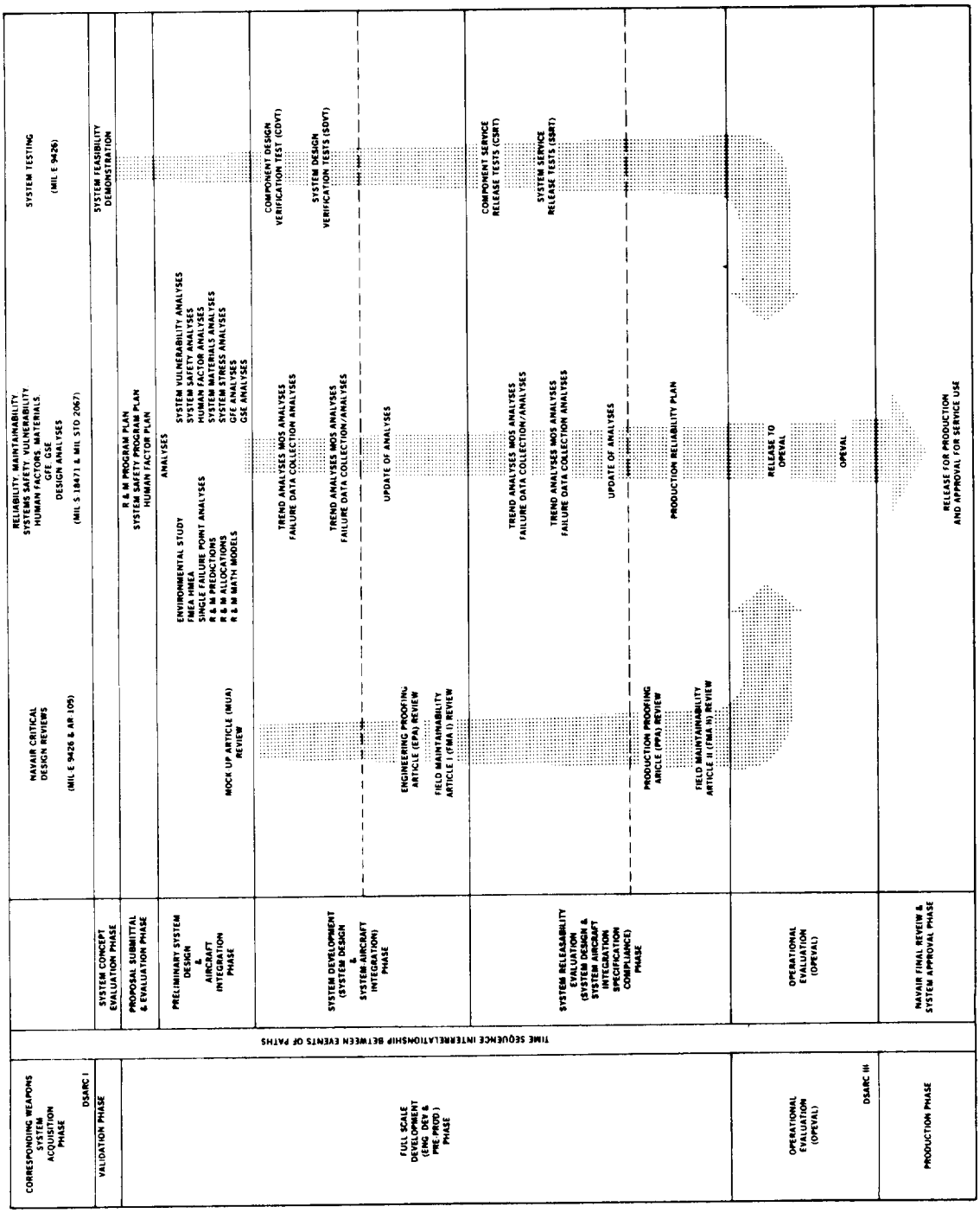


FIGURE 15.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A

TEST WITNESS GUIDELINES AND CHECKLIST

_____AIRPLANE

MODEL _____AIRCREW AUTOMATED ESCAPE SYSTEM

CONTRACT _____

WITNESS NAME _____ DATE _____

Ref: (a) NAVAIRSYSCOM LTR AIR-5312C/884: PDD of 28 Apr 1967 (Attached)

PART I: GENERAL INSTRUCTIONS

1. Before test:

a. Familiarize yourself with the test plan. _____

b. Familiarize yourself with the test site. _____

c. Obtain Polaroid/TV camera/Tape recorder and familiarize yourself with their operation. _____. Detailed manufacturer's instructions for operation of these articles will be left in the designated storage area for the articles. When relieving an on-site test witness, obtain a checkout on their operation prior to his departure.

d. Ensure supply of Polaroid film/TV tape/Recorder tape is adequate. (See 3m).

e. Discuss split of duties with partner. _____

2. During test preparations:

a. Observe preparations, checking off points on your checklist and verifying contractor use of approved checklist. Obtain a copy of the contractor's checklist.

b. Obtain copy of contractor list of reused components.

c. Obtain copy of contractor's list of escape system and dummy serial numbers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A. (continued)

d. Verify with contractor seat locations and dummy locations._____

e. Point out to contractor/test director any questionable items noted to obtain correction/explanation. Record description/photograph._____

WITNESS NAME _____ DATE _____

f. Insure test setup photographed prior to test initiation.

3. Post-test procedures:

a. With contractor representative and photographer, observe condition of escape system(s), dummy(ies), test equipment(s), sled(s) BEFORE OTHER PERSONNEL ARE GIVEN ACCESS TO THESE EQUIPMENTS._____

b. Insure photographs are taken of components as they rest, then examine for damage. If damage is found, record and photograph._____

c. Parts having significant damage are to be tagged and carefully moved to clean location for later disassembly. INSTRUCT PERSONNEL NOT TO JAR OR INITIATE DISASSEMBLY._____

d. Examine sleds for unusual damage. If found, record and photograph. If significant, arrange for it to remain undisturbed until determination can be made whether damage resulted from or caused escape system malfunctions/ problems.

e. Examine traces for any unusual characteristics. If found, note. _____

f. Prepare written brief for lead laboratory and NAVAIR project engineers and next Government witness concerning the test, the test procedures, test results, problem areas, etc.

g. Advise relief witness of any actions not completed which he may have to check out.

h. Describe for lead laboratory and NAVAIR project engineers any changes in procedures of this checklist which might improve the program._____

i. Pass on to relief Polaroid/TV camera/Tape recorder with films/TV tapes, recorder tape and operating instructions.

j. Advise NAVAIRTESTCEN project engineer (301-863-4141) if Polaroid, TV camera/Tape recorder malfunctioning.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

k. Unless permission to the contrary is granted, by lead laboratory/ NAVAIR project engineers, remain to brief relief witness on site so that any peculiarities may be described and relief witness gains some familiarity with test site and equipments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

WITNESS NAME_____DATE _____

1. Record quantity of film/TV tape used.
Polaroid _____ TV tape _____

m. Use the preprinted order form to replenish film and TV tape supplies (This form is stored with the film and TV Tape).

PART II - A TEST SITE PREPARATION (if provided space is insufficient, use back of these pages).

1. DATE _____

2. Test location: _____Altitude: _____ft.

3. Test plan:

a. Number and title: _____
_____
_____

b. Planned test speed: _____kt.

c. Government approval: (Activity and Date): _____
_____

4. Camera positions:

a. Comply with test plan? Yes/No.

b. Describe variations and contractor/testing activity reasons:_____
_____
_____
_____

5. Telemetry:

a. Calibration data (obtain copy).

b. Has contractor/testing activity conducted a checklist check out of telemetry: Yes/No.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

WITNESS NAME_____ DATE _____

(c) Comply with test plan? Yes/No.

(d) Data acquisition system malfunction(s) noted prior to test (Yes/No) and corrected (Yes/No.). If not corrected, give contractor/testing activity reasons.

_____

_____

_____

_____

(e) Describe nature of malfunctions (corrected or uncorrected):_____

_____

_____

_____

6. Personnel.

(a) Contractor test engineer/director: _____

(b) Testing activity/test engineer/director: _____

(c) Government witnesses (name and activity):

(1) _____

(2) _____

(3) _____

(d) Were personnel briefed concerning who was authorized to approach test equipment and/or test articles immediately after test? Yes/No.

7. Test postponement? Yes/No.

(a) If postponed, contractor/testing activity reasons:_____

_____

_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

WITNESS NAME _____ DATE _____

8. Meteorological data:

a. Wind: Force _____kt. Direction _____(steady/gusty). Tract direction _____

b. Clear/cloudy: approximate cloud coverage _____ tenths.

c. Temperature: Dry bulb _____. Wet bulb _____.

d. Barometric pressure: _____

PART II - B SLED/AIRCRAFT PREPARATION (Strike inapplicable)

1. Calibration data (obtain copy).

2. Contractor testing activity using checklist(s) in preparing sled/aircraft and escape system for test? Yes/No.

3. Contractor/testing activity verifying completion of each preparatory action? Yes/No. (Obtain copy of checklist).

4. Pusher sled:

a. Type and quantity of propulsion units: _____

_____

b. Comply with test plan? Yes/No.

c. Describe variations and contractor/testing activity reasons:

_____

_____

5. Cockpit/sled

Seat Ser #	Seat Location *	Seat Height Adjustment	Dummy Percentile/ Weight	Parachute Ser #

*LF - Left Front, RR - Right Rear, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

WITNESS NAME _____ DATE _____

a. Dummy limb retention used: _____

_____

b. Flight equipment/survival equipment used (list nomenclature and size):

		Cockpit Position*			
		LF	RF	LR	RR
(1)	Helmet				
(2)	Flight coveralls				
(3)	Pressure suit (Yes/No) Pressurized				
(4)	Exposure suit				
(5)	Anti-g suit				
(6)	SV-2 survival vest (ballasted?) Yes/No				
(7)	Torso suit				
(8)	Gloves				
(9)	Flight boots				
(10)	O_2 mask and hoses				
(11)	Flotation gear				
(12)	Survival kit (equipped or ballasted)				

c. Reuse of escape system components:

(1) Is there a Government approved list of components which may be reused? Yes/No.

(2) Name of activity giving approval and date of approval: _____

_____

(3) Component reuse list identification (number and title): _____

_____

(4) Parts reused in test (P/N , S/N, Nomenclature)

P/N	S/N	Nomenclature

*LF - Left front, RR - Right rear, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

WITNESS NAME _____ DATE _____

(5) Were parts inspected prior to reuse? Yes/No.

By whom and how?_____

_____

d. Dummy telemetry:

(1) Contractor/testing activity checked out:

*LF - Left Front, RR - Right Rear, etc.

(a) Obtain checklist and calibration data:

(b) Prior to dummy installation in sled. Yes/No.

(c) After dummy installation in sled. Yes/No.

(2) Malfunctions noted (Yes/No) and corrected (Yes/No). If not corrected, give contractor/testing activity reasons:_____

_____

_____

e. Pretest photographs taken to show test installation? Yes/No. If not, why not?_____

_____

_____

PART III POST TEST OBSERVATIONS

1. Any apparent malfunctions during test? Yes/No. Describe: _____

_____

_____

_____

2. Range control:

a. Any unauthorized personnel approach, touch, handle any test equipments and/or test articles before contractor test engineer/director, testing activity test engineer/director and Government witness gave permission? Yes/No.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX A (continued)

Names and actions: _____
_____
_____

3. Damage inspection and recording:

a. Noted in escape system or its components? Yes/No. Describe: _____
_____
_____
_____

b. Noted in test sled or pusher sled? Yes/No. Describe: _____
_____
_____
_____

c. Noted in GFE and/or flight equipment/survival equipment? Yes/No. Describe: _____
_____
_____
_____

d. Were photographs taken? Yes/No. Photographer's name/activity/telephone number: _____
_____
_____

4. Did any malfunctions occur in the data acquisition system? Yes/No. Describe: _____
_____
_____
_____

5. Refer to paragraph 3 POST TEST Procedures (page 83)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

APPENDIX B

GENERAL COMMENTS CONCERNING TESTING OF AIRCREW AUTOMATED ESCAPE SYSTEMS

Due to the expense in terms of money, manpower, hardware, and time involved in conducting aircrew automated escape systems tests; escape system developmental, design verification and service release test programs generally involve statistically small samples. As a result of the small sample sizes tested, individual successful tests statistically are almost insignificant; whereas individual failures, incipient failures/marginal successes, and other anomalies acquire considerable statistical significance. Accordingly, throughout an escape system test program, from the initial planning phase through the submittal and approval of the final report, all participants must direct their efforts towards the detection of such failures, incipient failures/marginal successes, and other anomalies indicative of system and/or component malperformance.

There are at least eight (8) phases of any test and/or test program and the actions of the test participants in each phase can contribute to the success or the failure of the vital data gathering and analyses and the malperformance detection functions. These eight phases are:

(1)	Pre-test planning	Define the test(s)/test program purpose(s), the configuration of the test articles and fixtures, the data to be collected, the data acquisition techniques and equipments, test site, test personnel and their responsibilities, safety procedures, and system operation and potential failure modes.
(2)	Fabrication of test articles and fixtures	Design and manufacture all test articles and test fixtures, including data acquisition subsystems, in the required configuration.
(3)	Test set-up	Prepare test articles within test fixtures, ensure records are current concerning condition and number of previous uses of each reusable part being used, and prepare test site and data acquisition subsystems for individual tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

(4)	Conducting test	Apply the required test conditions and/or conditioning and function the test article.
(5)	Post-test test site clean-up	Locate, record location and condition of, identify, and tag all pieces/debris from the test and furnish along with instrumentation records and films for analyses.
(6)	Post-test analyses	Evaluate instrumentation records, films and test site records (including comparison with similar data obtained from previous tests), and examine all expended/installed hardware and pieces and debris for indications of any failures, incipient failures/ marginal successes and other anomalies indicative of substandard performance, and develop rationale(s) concerning suspected cause(s) of such malperformance(s).
(7)	Post-test design analyses	Develop necessary remedial actions and fully define supporting rationale(s); and ascertain need for, and develop supporting rationale for, re-tests or special tests.
(8)	Test reporting	Summarize test purpose, test article configuration, test conditions, and test results; identify and reference all films, furnish copies of all failure analyses, and describe any design changes determined necessary with a brief supporting rationale.

Admittedly all phases of a test/test program are important, however several phases, in particular, often seem to be downgraded with very minimal effort expended in their execution. These are:

(1) Pre-test planning.
(2) post-test test site clean-up
(3) Post-test analyses.
(4) Post-test design analyses.

As a consequence most tests fail to yield their fullest potential as a valuable design or quality assurance tool in preparing an aircrew automated escape system for installation and service use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

A frequent additional consequence is program schedule slippage. It should be remembered that in the case of an escape system being perfected for delivery with a new weapons system, such delays may impact adversely the weapons system schedule. Even more important is the consequence of slippage in a retrofit situation. Since escape system retrofit only has occurred whenever performance can be improved measurably, such delays in completing the retrofit system can result in injuries/fatalities which could have been prevented by the on-time incorporation of the improved system.

The balance of this discussion attempts to outline briefly what is necessary to ensure the satisfactory completion of the above-listed test/test program phases.

I. Pre-test planning

One of the most important, yet often neglected, aspects of the pre-test planning phase is the careful, thorough definition of the test purpose/objective. The test purpose should be defined in clear, concise terms and, if a test is a part of a series of tests, the definition of its purpose should include a brief statement relating the individual test's purpose to the purpose of the test series.

Once the test purpose has been defined, and only then, the test article configuration, test fixture configuration, data collection requirements, data acquisition subsystem configuration, test site preparation, test personnel and their responsibilities, safety procedures, and test evaluation criteria can be defined meaningfully. The test purpose definition is the most vital single element in determining the criteria for judging test success or failure since a specific test might be conducted solely for the purpose of evaluating the performance and/or effects of a particular element of the system and not of the system itself. Specifically, a test of a particular element of an escape system could produce an acceptable system result, i.e. recovery of a dummy under a full parachute, and yet be considered a failure based upon the unacceptable performance of the particular element being tested. On the other hand, despite unacceptable system results, i.e. non-recovery of a dummy, a test of a particular element could be adjudged successful based upon the acceptable performance of the particular element being tested. In the latter event, post-test investigation should examine the test both to analyze those aspects directly related to the performance of the particular element being tested and, separately, those aspects leading to, or causing, unsatisfactory system performance.

II. Fabrication of test articles and fixtures

The primary objectives of conducting tests are:

> to ascertain whether the system (element) as designed and manufactured performs in an acceptable manner reliably.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

to ascertain the performance capabilities and limitations of the system (element) as designed and manufactured.

Therefore, it is vital to the success of each aircrew escape system test program and, more importantly, to the safety of the eventual users of the system, that every test article truly represent the ultimate production line articles. The test article configuration must be subject to the same strict controls following establishment of the test baseline configuration as will be imposed ultimately on the production line article configuration. Each test article and its components, to be valid test articles, must be procured, manufactured, assembled, and inspected to the same specifications and standards and with the same instructions, equipments, tools, fixtures, procedures, and processes as will be employed in the manufacture of each production line article. Any variation between the test article and production line article, particularly any increased care and attention given to the test article, makes that article non-representative and invalid and may make conclusions concerning system performance and reliability tenuous or invalid.

For the same reasons, the test fixtures must simulate closely, or if feasible actually duplicate exactly, the actual articles which they represent. Differences in geometry, structural response characteristics and/or aerodynamic characteristics between the test fixture(s) and the actual article(s) which they represent, can cause major variations between the system performance during testing and that of the in-service installed system.

In past aircrew escape system programs such differences between test articles and production line articles and/or between test fixtures and actual articles which they represent have resulted in both:

- the correction of apparent problems encountered during testing only to learn that the problems would not have existed in the in-service installation of production line aircrew escape systems.

- the post-test appearance of problems with in-service installations of production line aircrew escape systems although no indications of similar problems were revealed by testing.

Both situations generally involve considerable cost growth, schedule slippage and jeopardy of aircrew safety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

III. Test set-up

In keeping with the primary objectives of testing, it is important that test articles be mounted properly within the test fixtures and complete approved instrumentation be installed properly to measure and record system performance data. In preparation of the test article, test fixture and test site for a test considerable care is required to document fully:

- test article identification.
- test article re-use status.
- test article configuration.
- test site.
- test site conditions.
- test fixture identification.
- test fixture condition.
- instrumentation types and identification.
- instrumentation locations
- instrumentation calibration.
- test set-up crew.
- witnesses.
- personnel having access to the test set-up.
- test conditions.

To ensure the validity of the test and, oftentimes, the safety of personnel and property, access to the test site and particularly the test set-up must be controlled strictly. Failure to control access can result in inadvertent/premature operation of test articles and/or equipment, distraction of test set-up crew, and/or alteration of the test set-up, any of which can be hazardous to onlookers and test set-up crew and/or can jeopardize the test.

To ensure the reliable, safe, correct set-up of each test it is recommended that detailed set-up procedures and check lists be developed, verified in "dry run" set-ups, and used for each applicable subsequent test.

IV. Conducting test

If test preparations have been made correctly and test site access has been controlled strictly, the actual conduct of an aircrew escape system (element) test generally is straightforward. However, contingency plans should be available detailing procedures to be employed in the event the test does not occur exactly as planned, i.e. hangfire/misfire or system malfunction, to reduce the risk to the personnel, test set-up, test article, and property.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

V. Post-test test site clean-up

Trite as it may seem, good housekeeping is a vital and oft neglected aspect of test site management and of testing. Three important considerations are involved:

- maintaining full post-test test site access control,
- ensuring that all of the test article is located and returned for detailed analyses in conjunction with the analysis of other data, and
- precluding intermixing of pieces from various tests.

Frequently detailed analyses of markings and/or deformation of test article pieces/debris may confirm or explain data obtained from conventional data acquisition subsystems or may furnish the only clues so vital to ascertaining whether a part operated correctly and, if not, why not. On the other hand, when the test site is not cleaned following each test, the paternity of the bits and pieces found on site when attempting to piece together the cause(s) of a problem is in doubt and they contribute little but confusion and lost time to an investigation.

To facilitate the post-test location of test article elements and the subsequent evaluation of the test article, it is recommended that any ground cover be kept short so that parts can be located readily, a supply of marker wands (flags) be provided to assist recording the positions of parts relative to one another and to the test initiation point, and that the test crew be prepared to establish grids across the test site to organize any search for important missing parts.

VI. Post-test analyses

Very seldom does test data receive the thorough analysis or does the test article receive the thorough post-test examination necessary to extract all of the potentially useful information. In some instances, it has seemed that data was collected and subsequently treated only so as to furnish specification-required data formats.

All data should be treated and examined as though a problem had occurred during the test. This requires that the data be presented and examined in several different formats and that it be compared to data obtained from previous tests. All anomalies should be identified carefully and examined to ascertain their cause(s). Expended/used hardware and pieces/debris should be examined thoroughly both in the broad scope and in minute detail and all

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

markings/deformations noted and cataloged. Where pieces/components used in a test have been re-used and/or may be re-used the condition of recovered pieces/components should be compared to their previous condition and their post-test condition cataloged carefully in a log for future comparison. Cataloging such markings/deformations often is best accomplished by sketches and photographs. Failure to maintain a thorough log cataloging carefully the changes in part/component condition may introduce confusion as to the role a part/component had in a failure or as to why a particular part/component failed. An attempt should be made to ascertain their causes.

Graphs presenting trends should be employed extensively both for instrumentation record-derived data and for the frame-by-frame changes, be it position, attitude or shape, etc., of an escape system and/or its parts undergoing test.

Without careful thorough analyses of all data and all test article hardware either an incipient failure/marginal success or other anomaly may not be detected during the testing or a known problem may not receive sufficient analyses to permit the sound development of remedial action. Either circumstance can permit systems with inherent flaws to be delivered to the fleet for service use. Generally such buried inherent flaws then are detected only when an escape system fails under conditions permitting its relatively undamaged recovery and the failure results in fatal, serious or unusual injuries to the user. Unfortunately, many Navy ejections occur in locations where the equipment cannot be recovered for investigation and the cause(s) of injuries remains undetermined.

VII. Post-test design analyses

Following the completion of data analyses/hardware examination, the component and system designs often require analyses to ascertain how to correct a problem that has occurred without creating new problems. In every case, the decision to change or not to change a design, should be supported by a brief detailed rationale.

VIII. Test reporting

Test reports have a crucial role in testing since the best conducted tests serve little purpose if the information is not communicated to others involved and concerned. Poorly organized and/or prepared test reports often fail in their communications role. A test report serves to advise all affected/ concerned personnel/organizations as to the nature, purpose and results of a test and the reports are used as an aid in determining whether to approve a system, whether to impose limits upon a system, and, subsequent to approval and release, as an aid in studying reported in-service malfunctions and problems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

IX. Other aspects

One problem that arises often enough to be of concern is the care given to test articles to ensure that they will be successful. Sometimes this tender loving care distorts the test article configuration sufficiently to cause it to be non-representative of the actual system. This distortion can occur in at least two ways and generally is done with the best of intentions.

(1) test personnel make on-the-spot changes to correct what appear to be, or actually are, defects or problems. These changes often are not authorized by cognizant design personnel and, sadly, just as often are not reported to the cognizant design personnel who, therefore, may never learn of the problem that was so corrected. Under these conditions, it frequently is impossible in later years to understand why a problem "just cropped up."

(2) A particularly insidious type of tender loving care that occurs is the 100 per cent inspection of many components used in the test articles even though those same components used in production articles undergo only a sampling per MIL-STD-105 or MIL-STD-414. Immediately the expected test results must be considered to be weighted more towards success than is the case for production articles.

It would seem that the appropriate approach to the problem would be to conduct normal sampling and record the results - lot accepted or lot rejected. In the case of an accepted lot, the lot only should be subjected to 100 per cent inspection to ensure full knowledge concerning characteristics of critical pieces/components so as to assist in the subsequent test data analyses hardware examinations. In the event the 100 per cent inspection should detect an unacceptable piece/component to be present in an accepted lot, then attention should be focused upon manufacturing, handling and quality assurance processes/procedures to ensure their correction and prevention of future occurrences.

It should be borne actively in mind by all test program participants that one important objective of escape system testing is to verify manufacturing, handling, and quality assurance processes/procedures as well as the design itself. Accordingly, actions which change the system on-site can be tolerated only when fully coordinated with cognizant design personnel who then make the appropriate changes to the design or to the manufacturing, handling and quality assurance procedures. During service release test such changes require the Naval Air Systems Command representative's approval before they may be effected.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:54Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF PARAGRAPHS CITING SPECIFICATIONS BY SPECIFICATION NUMBER

REFERENCED SPECIFICATIONS NUMBER	TITLE	PARAGRAPH NUMBER	SUBJECT	PAGE
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for	3.8.1	Component/subsystem service release/ qualification tests	22
		3.8.3.1	Environmental test regimen	27
MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for	1.1	Scope	1
		3.8.2.5	Demonstration of aircrew capability to reach and actuate aircraft controls	26
		3.8.2.6	Demonstration of aircrew capability to reach and operate escape system controls	27
		3.8.2.7	Demonstration of aircrewmen percentile range accomodation	27
		3.8.3.1	Environmental test regimen	28
		3.8.4.3.2.5.1	Canopy implosion tests	32
		3.8.4.3.4.2	Purpose of tests	38
		3.8.4.5	Impulse noise levels	39
		3.9	Comparison of tested escape system's performance against performance requirements	40
		3.10	Prerequisits for commencing service release test (SRT)	40
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices	3.8.1.1	Cartridges	23
		3.8.1.7.3	Retardation and stabilization subsystem ballistic elements	24
		3.8.1.10	Escape path clearance and canopy removal subsystem	25
MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System; General Specification for	1.1	Scope	1
		3.8.2.5	Demonstration of aircrew capability to reach and actuate aircraft controls	26
		3.8.2.6	Demonstration of aircrew capability to reach and actuate escape system controls	27
		3.8.2.7	Demonstration of aircrewmen percentile range accomodation	27
		3.8.4.6	Cabin contamination tests	39

INDEX OF PARAGRAPHS CITING SPECIFICATIONS BY SPECIFICATION NUMBER

REFERENCED SPECIFICATIONS NUMBER	TITLE	PARAGRAPH NUMBER	SUBJECT	PAGE
		3.8.3.1	Environmental test regimen	28
		3.8.4.3.2.5.1	Canopy implosion tests	32
		3.8.4.3.4.2	Purpose of tests	38
		3.8.4.5	Impulse noise levels	39
		3.8.4.7	Flotation tests	39
		3.9	Comparison of tested escape system's performance against performance requirements	40
		3.10	Prerequisites for commencing service release test (SRT)	40
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices	3.8.1.2	Cartridge Actuated Devices	23
		3.8.1.7.3	Retardation and stabilization subsystem ballistic elements	24
		3.8.1.10	Escape path clearance and canopy removal subsystem	25
MIL-S-81018	Survival Container, Aircraft Seat With Oxygen, General Specification for	3.8.1.9	Survival Equipment Containers	25
MIL-S-81040	Survival Kit Containers, Aircraft Without Oxygen, General Specification for	3.8.1.9	Survival Equipment Containers	25
MIL-D-81514	Device, Restraint Harness Take-up, Inertia-locking, Powered-Retracting; General Specification for	3.8.1.5	Ballistically-powered shoulder harness take-up device	23
MIL-A-81573	Aircrew Escape System Descriptive and Performance Data Presentations, Requirements and Formats for;	3.8.4.3.4.2	Purpose of tests	38

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF PARAGRAPHS CITING SPECIFICATIONS BY SPECIFICATION NUMBER

REFERENCED SPECIFICATIONS NUMBER	TITLE	PARAGRAPH NUMBER	SUBJECT	PAGE
MIL-C-81590	Cockpit Canopy System, Fixed Wing Single and Multiplace Fighter, Attack, and Trainer Aircraft; General Specification for	3.8.1.10	Escape path clearance and canopy removal subsystem	25
		3.8.4.3.2.5.1	Canopy implosion tests	32
MIL-A-81973	Aircrew Automated Escape Systems, Test Vehicles for Ground Track Performance Testing, Rocket and/or Jet Engine Propelled; General Specification for	3.2	Test Vehicle for performance demonstration	6
MIL-D-81980	Design and Evaluation of Signal Transmission, Subsystems; General Specification for	3.8.1.4.1	Ballistic signal transmission line testing	23
MIL-A-85041	Aircrew Escape Propulsion Systems, General Specification for	3.8.1.3	Rocket motor/rocket catapult propulsion subsystem and/or devices	23
		3.8.1.7.2	Rocket motor type decelerators/ stabilizers	24
AR 46	HERO (Hazards of Electromagnetic Radiation to Ordnance) Requirements for HERO tests, Analysis Documentation	3.8.3.4	Electromagnetic compatibility tests	29
AR 72	Aircrew Automated Escape System, Service Release Test Programs, Contractor Furnished Equipments and Government Furnished Equipments for	3.1.3	Contractor/government furnished equipment requirements	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF PARAGRAPHS CITING SPECIFICATIONS BY SPECIFICATION NUMBER

REFERENCED SPECIFICATIONS NUMBER	TITLE	PARAGRAPH NUMBER	SUBJECT	PAGE
AR 105	Aircrew Automated Escape System I-Process Design Conformance Inspection Articles, Standard Procedures for Evaluation of;Requirements for	3.10.1	Mock-up proofing and field maintainability review article	41
FED-STD-191	Textile Test Methods	3.8.1.13.1	Flame resistance	25
MIL-STD-442	Aerospace Telemetry Standards	3.4.1	Electronic instrumentation	8
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment	3.8.3.4	Electromagnetic compatibility tests	29
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of	3.8.3.4	Electromagnetic compatibility tests	29
MIL-STD-810	Environmental Test Methods	3.8.3.1	Environmental test regimen	27
MIL-STD-831	Test Reports, Preparation of	3.6 3.6.5	Reports of tests Final test report	16 19
MIL-STD-858	Testing Standard for Personnel Parachutes	3.8.1.6 3.8.1.7.1(a)	Recovery parachute subsystem aerodynamic decelerators/stabilizers	23 23
MIL-STD-882	System Safety Program for systems and Associated Subsystems and Equipment, Requirements for	4.2.1.1 (c) (15)	No test	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF PARAGRAPHS CITING SPECIFICATIONS BY SPECIFICATION NUMBER

REFERENCED SPECIFICATIONS NUMBER	TITLE	PARAGRAPH NUMBER	SUBJECT	PAGE
MIL-STD-888	Testing Program for Emergency Escape Capsule Recovery Systems	3.8.1.6	Subsystem Recovery parachute	23
MIL-STD-961	Outline of Forms and Instructions for the Preparation of Specifications and Associated Documents	3.8.1	Component/Subsystem service release/ qualification tests.	22
MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for	3.4	Data acquisition	7
		3.4.4.1	Test data analysis requirements	15
		3.8.1	Component/Subsystem Service release/ qualification tests	23
		3.8.4.3.4	Performance tests	36
		3.8.4.3.4.1.1	Test program modifications	37
		3.8.4.8	Field maintainability verification	39
		3.8.5	Maintainability and reliability testing	40
		3.8.5.1	Installed maintenance-free life demonstration	40

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
1.1	Scope	1	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.1.3	Contractor/government furnished equipment requirements	6	AR-72	Aircrew Automated Escape System, Service Release Test Programs, Contractor Furnished Equipments and Government Furnished Equipments; Requirements for
3.2	Test vehicle for performance demonstrations	6	MIL-A-81973	Aircrew Automated Escape Systems, Test Vehicles for Ground Track Performance Testing, Rocket and/or Jet Engine Propelled; General Specification for
3.4	Data acquisition	7	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for
3.4.1	Electronic instrumentation	8	MIL-STD-442	Aerospace Telemetry Standards
3.4.4.1	Test data analysis requirements	15	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for
3.6	Reports of tests	16	MIL-STD-831	Test Reports, Preparation of
3.6.5	Final test report	19	MIL-STD-831	Test Reports, Preparation of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
			AR 47	Technical Information Requirements for Automated Aircrew Escape Systems Proposal
3.8.1	Component/subsystem service release/qualification tests	22	MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment; General Specification for
		22	MIL-STD-961	Outline of Forms and Instructions for the Preparation of Specifications and Associated Documents
		23	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for
3.8.1.1	Cartridges	23	MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
3.8.1.2	Cartridge actuated devices	23	MIL-D-23615	Design and Evaluation of Cartridges for Cartridge Actuated Devices
3.8.1.3	Rocket motor/rocket catapult propulsion subsystem and/or devices	23	MIL-A-85041	Aircrew Escape Propulsion Systems, General Specifications for
3.8.1.4.1	Ballistic signal transmission line testing	23	MIL-D-81980	Design and Evaluation of Signal Transmission Subsystems; General Specification for
3.8.1.5	Ballistically-powered shoulder harness take-up device	23	MIL-D-81514	Device, Restraint Harness Take-up, Inertia-Locking, Powered-Retracting; General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
3.8.1.6	Main recovery parachute subsystem	23	MIL-STD-858	Testing Standard For Personnel Parachutes
			MIL-STD-888	Testing Program for Emergency Escape Capsule Recovery Systems
3.8.1.7.1a	Aerodynamic decelerators/stabilizers	23	MIL-STD-858	Testing Standard for Personnel Parachutes
3.8.1.7.2	Rocket motor type decelerators/ stabilizers	24	MIL-A-85041	Aircrew Escape Propulsion Systems, General Specification for
3.8.1.7.3	Retardation and stabilization Subsystem ballistic elements	24	MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
			MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
3.8.1.9	Survival equipment containers	25	MIL-S-81018	Survival Container, Aircraft Seat with Oxygen; General Specification for
			MIL-S-81040	Survival Kit Containers, Aircraft Seat, Without Oxygen; General Specification for
3.8.1.10	Escape path clearance and canopy removal subsystem	25	MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
			MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
			MIL-C-81590	Cockpit Canopy System, Fixed Wing Single and Multiplace Fighter, Attack, and Trainer Aircraft; General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
3.8.1.13.1	Flame resistance	25	FED-STD-191	Textile Test Methods
3.8.2.4	Demonstration of Aircrew capability to reach and actuate aircraft controls	26	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type; General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System; General Specification for
3.8.2.6	Demonstration of aircrew capability to reach and actuate escape system controls	27	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type; General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule Capsule System; General Specification for
3.8.2.7	Demonstration of aircrewmen percentile range accommodation	27	MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
		27	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type; General Specification for
3.8.3.1	Environmental test regimen	27	MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment; General Specification for
			MIL-STD-810	Environmental Test Methods
		28	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type; General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.8.3.4	Electromagnetic compatibility tests	29	MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
			MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
			AR 46	HERO (Hazards of Electromagnetic Radiation to Ordnance) Requirements for HERO Tests, Analysis Documentation
3.8.4.3.2.5.1	Canopy implosion tests	32	MIL-C-81590	Cockpit Canopy System, Fixed Wing Single and Multiplace Fighter, Attack and Trainer Aircraft, General Specification for
			MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type; General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.8.4.3.4	Performance tests	36	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program Requirements for
3.8.4.3.4.1.1	Test program modifications	37	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program Requirements for
3.8.4.3.4.2	Purpose of tests	38	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for
			MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program Requirements for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
			MIL-A-81573	Aircrew Escape System Descriptive and Performance Data Presentations, Requirements and Formats for; General Specification for
3.8.4.5	Impulse noise levels	39	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.8.4.6	Cabin air contaminantion tests	39	MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.8.4.7	Flotation tests	39	MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.8.4.8	Field Maintainability verification	39	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for
3.8.5	Maintainability and reliability testing	40	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for
3.8.5.1	Installed maintenance-free life demonstration	40	MIL-STD-2067	Aircrew Automated Escape Systems, Reliability and Maintainability (R/M) Program; Requirements for

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX OF CITED SPECIFICATIONS BY CITING PARAGRAPH

PARAGRAPH NUMBER	SUBJECT	PAGE	REFERENCED SPECIFICATIONS NUMBER	TITLE
3.9	Comparison of tested escape system's performance against performance requirements	40	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
			MIL-A-81573	Aircrew Escape System Descriptive and Performance Data Presentations, Requirements and Formats for; General Specification for
3.10	Prerequisites for commencing service release test (SRT)	40	MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type, General Specification for
			MIL-STD-2067	Aircrew Automated Escape System, Reliability and Maintainability (R/M) Program; Requirements for
			MIL-A-23121	Aircrew Environmental Escape and Survival Cockpit Capsule System, General Specification for
3.10.1	Mock-up proofing and field maintainability review article	41	AR 105	Aircrew Automated Escape System In-process Design Conformance Inspection Articles, Standard Procedures for Evaluation of; Requirements for
4.2(c)(15)	Criteria for judging the acceptability of the results of individual tests and test programs	45	MIL-STD-882	System Safety Program for Systems and and Associated Subsystems and Equipment, Requirements for

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
APPENDIX A TEST WITNESS GUIDELINES AND CHECKLIST		82
APPENDIX B GENERAL COMMENTS CONCERNING TESTING OF AIRCREW AUTOMATED ESCAPE SYSTEMS		91
APPLICABLE DOCUMENTS	2	2
ISSUES OF DOCUMENTS	2.1	2
FIGURE 1 Standard escape trajectory plot		63
FIGURE 2 Sample canopy damage chart (for 24 and 28 foot canopies)		64
FIGURE 3 Sample canopy damage chart		65
FIGURE 4 Sample dummy damage chart		66
FIGURE 5 Dummy damage chart		67
FIGURE 6 Sample equipment damage chart		69
FIGURE 7 Equipment damage chart		70
FIGURE 8 Escape system performance, aircraft pitch angle vs. terrain clearance altitude for safe escape		74
FIGURE 9 Escape system performance, craft bank angle vs. terrain clearance altitude for safe escape		75
FIGURE 10 Aircrew automated escape system performance; aircraft bank angle, sink rate and airspeed vs. terrain clearance required for safe escape		76
FIGURE 11 Aircrew automated escape system; aircraft bank angle, dive angle and airspeed vs. terrain clearance required for safe escape		77
FIGURE 12 In-process design conformance inspection for escape system		78
FIGURE 13 Sample acceleration trace depicting required technique for measuring peak acceleration		79

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

FIGURE 14 Sample acceleration trace depicting required technique for measuring highest rate of G onset 80

FIGURE 15 Aircrew automated escape system design evaluation program event flow sequence diagram 81

NOTES

FUNCTION OF DATA REQUIREMENTS 6.2 48

INTENDED USE 6.1 48

Data Requirements 6.2.1 49

PACKAGING 5 48

QUALITY ASSURANCE PROVISIONS 4 41

CRITERIA FOR JUDGING THE ACCEPTABILITY OF RESULTS OF INDIVIDUAL TESTS AND TEST PROGRAMS 4.2 41

RESPONSIBILITY FOR INSPECTION 4.1 41

Sled Inspections 4.1.1 41

REQUIREMENTS 3. 4

ACCEPTANCE OF SYSTEMS 3.7 20

COMPARISON OF TESTED ESCAPE SYSTEM'S PERFORMANCE AGAINST PERFORMANCE REQUIREMENTS 3.9 40

COMPONENTS 3.1 4

Test Articles 3.1.1 5

Test Article Configuration changes 3.1.1.1 5

Test Article Reuse 3.1.2 5

Capsule Type Escape Systems 3.1.2.1 5

Ejection Seat Type Escape Systems 3.1.2.2 6

Contractor/Government Furnished Equipment Requirements 3.1.3 6

DATA ACQUISITION 3.4 7

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

Electronic Instrumentation	3.4.1	8
Structure-mounted Instruments	3.4.1.1	8
Dummy-Mounted Instruments	3.4.1.2	8
Impulse Noise Level Instrumentation	3.4.1.3	9
Photographic Requirements	3.4.2	9
High Speech Photographic Coverage	3.4.2.1	10
Pre-test Photographs	3.4.2.2	10
Post-test Photographs	3.4.2.3	10
Other Photographic Coverage	3.4.2.4	11
Test Data Requirements	3.4.4	11
Test Data Analysis Requirements	3.4.4.1	15
Timing Correlation	3.4.3	11
GENERAL PROBLEM AREAS	3.12	41
GOVERNMENT WITNESSES	3.11	41
PREREQUISITES FOR COMMENCING SERVICE RELEASE TEST (SRT) PROGRAM	3.10	40
Mock-up, Proofing and Field Maintainability Review Article Evaluations	3.10.1	41
REPORTS OF TESTS	3.6	16
Design Verification Test (DVT) Reports	3.6.4	18
DVT Summarization Report	3.6.4.2	19
Individual Test/Test Series Reports	3.6.4.1	18
Final Test Report	3.6.5	19
Test Data Presentation	3.6.1	16
Preliminary Report of Test Results	3.6.2	17
Individual Test Reports	3.6.3	18

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

TEST FACILITIES	3.3	6
Apparatus	3.3.1	6
Test Equipment	3.3.1.1	6
Test Instrumentation	3.3.1.2	6
Frequency Allocation	3.3.4	7
Test Area	3.3.2	7
Test Area Access Control	3.3.3	7
TEST PLAN	3.5	15
Test Plan Change Proposals	3.5.1	16
Government Facility Test Support Document	3.5.2	16
TEST VEHICLE FOR PERFORMANCE DEMONSTRATIONS	3.2	6
TYPES OF SERVICE RELEASE DEMONSTRATIONS AND TESTS	3.8	20
Aircrew Station Fit and Compatibility Tests	3.8.2	26
Demonstration of Aircrew Capability to Reach and Actuate Aircraft Controls	3.8.2.5	26
Demonstration of Aircrew Capability to Reach and Actuate Escape System Controls	3.8.2.6	27
Demonstration of Aircrewman Percentile Range Accommodation	3.8.2.7	27
Field of Vision	3.8.2.8	27
Manual Egress Demonstrations	3.8.2.4	26
Mock-up Role in Aircraft Fit and Compatibility Tests	3.8.2.1	26
Engineering Proofing Article Role in Aircraft Fit and Compatibility Tests	3.8.2.2	26
Production Proofing Article Role in Aircraft Fit and Compatibility Tests	3.8.2.3	26

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

Component/Subsystem Service Release/Qualification Tests	3.8.1	22
Signal Transmission Subsystem Testing	3.8.1.4	23
Ballistic Signal Transmission Line Testing	3.8.1.4.1	23
Ballistically Powered Shoulder Harness Take-up Device	3.8.1.5	23
Cartridges	3.8.1.1	23
Cartridge Actuated Devices	3.8.1.2	23
Escape Parameters Sensing and Escape System Mode of Operation Control Subsystem	3.8.1.8	24
Escape Path Clearance and Canopy Removal Subsystem	3.8.1.10	25
Escape System Controls	3.8.1.11	25
Recovery Parachute Subsystem	3.8.1.6	23
Personnel Restraint	3.8.1.12	25
Retardation and Stabilization Subsystem	3.8.1.7	23
Aerodynamic Decelerators/Stabilizers	3.8.1.7.1	23
Phased Aerodynamic Decelerators/Stabilizers	3.8.1.7.1.1	24
Rocket Motor Type Decelerators/Stabilizers	3.8.1.7.2	24
Retardation and Stabilization Subsystem Ballistic Elements	3.8.1.7.3	24
Rocket Motor/Rocket Catapault Propulsion and/or devices	3.8.1.3	23
Survival Equipment Containers	3.8.1.9	25
Upholstery, Covers and Cushions	3.8.1.13	25
Flame Resistance	3.8.1.13.1	25
Design Compliance Tests	3.8.4	29
Cabin Air Contamination Tests	3.8.4.6	39
Crash Loads Strength Demonstration	3.8.4.1	29

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:54Z
Check the source to verify that this is the currentversion before use.

Ground, Track and Flight Performance Demonstrations	3.8.4.3	30
Definitions	3.8.4.3.2	30
Crash Condition Escape Tests	3.8.4.3.2.4	31
Flight Tests	3.8.4.3.2.3	31
System Zero Speed and Low Speed Operational Tests	3.8.4.3.2.1	31
Track Tests	3.8.4.3.2.2	31
Underwater Escape Capability Tests	3.8.4.3.2.5	31
Canopy Implosion Tests	3.8.4.3.2.5.1	32
Manual Underwater Escape Capability Tests	3.8.4.3.2.5.2	32
Performance Tests	3.8.4.3.4	36
Performance Test Program	3.8.4.3.4.1	37
Test Program Modifications	3.8.4.3.4.1.1	37
Test Program Schedule	3.8.4.3.4.1.2	38
Purpose of Tests	3.8.4.3.4.2	38
Prerequisite for Testing	3.8.4.3.1	30
Procedures	3.8.4.3.3	33
Escape Path Clearance Tests	3.8.4.3.3.1	33
Ejected Weight	3.8.4.3.3.3	33
Ballasting	3.8.4.3.3.3.1	34
Configuration	3.8.4.3.3.5	34
Escape System Initiation	3.8.4.3.3.4	34
Ejection Seat Recovery Equipment	3.8.4.3.3.6	35
Instrumentation (recording and evaluating data)	3.8.4.3.3.7	35
Hard Line Data Transmission	3.8.4.3.3.7.1	36

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

Through the Canopy, Canopy Removal by Seat and Through Thermal/Radiation Closure Tests	3.8.4.3.3.2	33
Special Data Acquisition/Analyses	3.8.4.3.3.2.3	33
Through the Canopy and Canopy Removal by Seat Test	3.8.4.3.3.2.1	33
Through Thermal/radiation Closure Tests	3.8.4.3.3.2.2	33
Environmental Conditioning and Windblast Testing	3.8.3	27
Electromagnetic Compatibility Tests	3.8.3.4	29
Environmental Test Regimen	3.8.3.1	27
Explosive Decompression Tests	3.8.3.3	28
Windblast Loads Tests	3.8.3.2	28
Field Maintainability Verification	3.8.4.8	39
Flotation Tests	3.8.4.7	39
Impulse Noise Levels	3.8.4.5	39
Underwater Performance Demonstrations	3.8.4.4	38
Seat/Man Restraint Physiological Acceptability Demonstrations	3.8.4.2	29
Maintainability and Reliability Testing	3.8.5	40
Installed Maintenance-free Life Demonstration	3.8.5.1	40
SCOPE	1	1
CLASSIFICATION	1.2	1
TABLE I Seat/Man Restraint Physiological Acceptability Tower Test Program		51
TABLE II Track Test Program (Single Crew Station System)		54
TABLE III Track Test Program (Multi-Crew Station Systems and Cockpit Capsule Systems)		55
TABLE IV Flight Test Program (Single Seat Systems)		56

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.

TABLE V Crash Condition Escape Program 57

TABLE VI Underwater Escape Tests 58

TABLE VII Test Report Format 60

☆U.S. GOVERNMENT PRINTING OFFICE: 1979-603-022/108

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:54Z
Check the source to verify that this is the current version before use.